1 Problem definition

One dimensional Lagrangian rad-hydro equation in the diffusion approximation:

$$\frac{\partial v}{\partial t} - \frac{\partial u}{\partial m} = 0 \tag{1}$$

$$\frac{\partial u}{\partial t} + \frac{\partial p}{\partial m} = 0 \tag{2}$$

$$\frac{\partial e}{\partial t} + p\frac{\partial v}{\partial t} = -\frac{\partial s}{\partial m} \tag{3}$$

where the radiation energy flux (energy per unit area per unit time):

$$s\left(m,t\right) = -\frac{c}{3\kappa_R}\frac{\partial}{\partial m}\left(aT^4\right) \tag{4}$$

since the Diffusion coefficient is

$$D = \frac{c}{3\kappa_R \rho} \tag{5}$$

where κ_R is the Rosseland opacity.
Assuming an ideal gas law:

$$pv = \left(\gamma - 1\right)e = re \tag{6}$$

And a power law Rosseland opacity:

$$\frac{1}{\kappa_R\left(T,\rho\right)} = gT^{\alpha}\rho^{-\lambda} = gT^{\alpha}v^{\lambda} \tag{7}$$

a power law specific interal energy:

$$e\left(T,\rho\right) = fT^{\beta}\rho^{-\mu} = fT^{\beta}v^{\mu} \tag{8}$$

With Boundary conditions:

$$T\left(0,t\right) = T_0 t^{\tau} \tag{9}$$

$$p\left(0,t\right) = 0 \tag{10}$$

now we have:

$$e = \frac{pv}{r} = fT^{\beta}v^{\mu} \tag{11}$$

hence:

$$T = \left(\frac{pv^{1-\mu}}{rf}\right)^{\frac{1}{\beta}} \tag{12}$$

hence the flux is:

$$
\begin{aligned}
-s\left(m,t\right) &= \frac{c}{3\kappa_R}\frac{\partial}{\partial m}\left(aT^4\right) && (13)\\
&= \frac{c}{3}gT^{\alpha}v^{\lambda}\frac{\partial}{\partial m}\left(aT^4\right) && (14)\\
&= \frac{4c}{3\left(4+\alpha\right)}gv^{\lambda}\frac{\partial}{\partial m}\left(aT^{4+\alpha}\right) && (15)\\
&= \frac{4ac}{3\left(4+\alpha\right)}gv^{\lambda}\frac{\partial}{\partial m}\left(T^{4+\alpha}\right) && (16)\\
&= \frac{4ac}{3\left(4+\alpha\right)}gv^{\lambda}\frac{\partial}{\partial m}\left(\left(\frac{pv^{1-\mu}}{rf}\right)^{\frac{4+\alpha}{\beta}}\right) && (17)\\
&= \frac{4ac}{3\left(4+\alpha\right)\left(rf\right)^{\frac{4+\alpha}{\beta}}}gv^{\lambda}\frac{\partial}{\partial m}\left[\left(pv^{1-\mu}\right)^{\frac{4+\alpha}{\beta}}\right] && (18)\\
&= \frac{16\sigma}{3\left(4+\alpha\right)\left(rf\right)^{\frac{4+\alpha}{\beta}}}gv^{\lambda}\frac{\partial}{\partial m}\left[\left(pv^{1-\mu}\right)^{\frac{4+\alpha}{\beta}}\right] && (19)\\
&= \frac{16\sigma}{3\beta\left(rf\right)^{\frac{4+\alpha}{\beta}}}gv^{\lambda}\left(pv^{1-\mu}\right)^{\frac{4+\alpha-\beta}{\beta}}\frac{\partial}{\partial m}\left(pv^{1-\mu}\right) && (20)\\
&= Bv^{\lambda}\left(pv^{1-\mu}\right)^{\frac{4+\alpha-\beta}{\beta}}\frac{\partial}{\partial m}\left(pv^{1-\mu}\right) && (21)\\
& && (22)
\end{aligned}
$$

where, we used $a=\frac{4\sigma}{c}$ and, unlike the S-H article, we redfine:

$$B=\frac{16\sigma g}{3\beta\left(rf\right)^{\frac{4+\alpha}{\beta}}} \tag{23}$$

since the radiation constant and Stephan-Boltzman constant are related via: finally, the equation is:

$$\frac{\partial e}{\partial t}+p\frac{\partial v}{\partial t}=-\frac{\partial s}{\partial m} \tag{24}$$

$$s\left(m,t\right)=-\left(\frac{\beta}{4+\alpha}\right)B\left(v^{\lambda}\frac{\partial}{\partial m}\left(pv^{1-\mu}\right)^{\frac{4+\alpha}{\beta}}\right) \tag{25}$$

this form is convinient since when using self-similar profiles, the $\left(\frac{4+\alpha}{\beta}\right)$ term will cancel due to the $\left(pv^{1-\mu}\right)^{\frac{4+\alpha}{\beta}}$ power:

$$s\left(m,t\right)=-Bv^{\lambda}\left(pv^{1-\mu}\right)^{\frac{4+\alpha-\beta}{\beta}}\frac{\partial}{\partial m}\left(pv^{1-\mu}\right) \tag{26}$$

we also define:

$$n=\frac{4+\alpha}{\beta} \tag{27}$$

$$k = 1 - \mu \tag{28}$$

$$s(m,t) = -Bv^{\lambda}\left(pv^{k}\right)^{n-1}\frac{\partial}{\partial m}\left(pv^{k}\right) \tag{29}$$

the boundary condition is

$$T(0,t) = T_0 t^{\tau} \tag{30}$$

$$T(0,t) = \left(\frac{p(0,t)\,v^{1-\mu}(0,t)}{rf}\right)^{\frac{1}{\beta}} = T_0 t^{\tau} \tag{31}$$

$$p(0,t)\,v^{1-\mu}(0,t) = rf\left(T_0 t^{\tau}\right)^{\beta} = rfT_0^{\beta}t^{\tau\beta} = \left((rf)^{\frac{1}{\beta}}\,T_0\right)^{\beta} t^{\tau\beta} = A^{\beta}t^{\tau\beta} \tag{32}$$

$$A = T_0\,(rf)^{\frac{1}{\beta}} \tag{33}$$

hence the problem is defined as:

$$\frac{\partial v}{\partial t}-\frac{\partial u}{\partial m}=0 \tag{34}$$

$$\frac{\partial u}{\partial t}+\frac{\partial p}{\partial m}=0 \tag{35}$$

$$\frac{\partial}{\partial t}\left(\frac{pv}{r}\right)+p\frac{\partial v}{\partial t}=-\frac{\partial s}{\partial m} \tag{36}$$

where the flux is:

$$s\left(m,t\right)=-Bv^{\lambda}\left(pv^{k}\right)^{n-1}\frac{\partial}{\partial m}\left(pv^{k}\right) \tag{37}$$

$$n=\frac{4+\alpha}{\beta},\ k=1-\mu \tag{38}$$

with the boundary condition:

$$p\left(0,t\right)v^{1-\mu}\left(0,t\right)=A^{\beta}t^{\tau\beta} \tag{39}$$

$$p\left(0,t\right)=0 \tag{40}$$

hence the problem is characterized by:
the dimensional parameters:

$$A=T_0\left(rf\right)^{\frac{1}{\beta}},\ B=\frac{16\sigma g}{3\beta\left(rf\right)^{\frac{4+\alpha}{\beta}}} \tag{41}$$

and independent dimensional variables:

$$m,\ t \tag{42}$$

and the dependent dimensional profiles:

$$v\left(m,t\right),\ u\left(m,t\right),\ p\left(m,t\right) \tag{43}$$

1.1 Self similar solution

Units:

v	u	p	A	B	m	t
$M^{-1}L$	LT^{-1}	MLT^{-2}	?	?	M	T

To find $[A]$:

$$p\left(0,t\right)v^{1-\mu}\left(0,t\right)=A^{\beta}t^{\tau\beta} \tag{44}$$

$$\left[pv^{1-\mu}\right]=\left[A\right]^{\beta}\left[T\right]^{\tau\beta} \tag{45}$$

$$\left[MLT^{-2}M^{-1+\mu}L^{1-\mu}\right]=\left[A\right]^{\beta}\left[T\right]^{\tau\beta} \tag{46}$$

$$[A]=\left[MLT^{-2}M^{-1+\mu}L^{1-\mu}\right]^{\frac{1}{\beta}}[T]^{-\tau} \tag{47}$$

mass power:

$$\frac{1-1+\mu}{\beta}=\frac{\mu}{\beta} \tag{48}$$

length power:

$$\frac{1+1-\mu}{\beta}=\frac{2-\mu}{\beta} \tag{49}$$

time power:

$$\frac{-2}{\beta}-\tau \tag{50}$$

hence:

$$[A]=[M]^{\frac{\mu}{\beta}}[L]^{\frac{2-\mu}{\beta}}[T]^{-\frac{2}{\beta}-\tau} \tag{51}$$

To find $[B]$:

$$\frac{\partial e}{\partial t}+p\frac{\partial v}{\partial t}=\left(\frac{\beta}{4+\alpha}\right)B\frac{\partial}{\partial m}\left(v^{\lambda}\frac{\partial}{\partial m}\left(pv^{1-\mu}\right)^{\frac{4+\alpha}{\beta}}\right) \tag{52}$$

$$\left[\frac{\text{energy}}{\text{mass}\times\text{time}}\right]=[B]\left[\frac{\partial}{\partial m}\left(v^{\lambda}\frac{\partial}{\partial m}\left(pv^{1-\mu}\right)^{\frac{4+\alpha}{\beta}}\right)\right] \tag{53}$$

$$\left[L^2T^{-3}\right]=\frac{[B]}{[M^2]}\left[M^{-1}L\right]^{\lambda}\left[MLT^{-2}\right]^{\frac{4+\alpha}{\beta}}\left[M^{-1}L\right]^{(1-\mu)\frac{(4+\alpha)}{\beta}} \tag{54}$$

isolating the units of B, we get for the mass:

$$2+\lambda-\frac{4+\alpha}{\beta}+(1-\mu)\frac{(4+\alpha)}{\beta}=2+\lambda-\mu\frac{(4+\alpha)}{\beta}=\frac{\beta(2+\lambda)-\mu(4+\alpha)}{\beta} \tag{55}$$

for the length:

$$2-\lambda-\frac{4+\alpha}{\beta}-(1-\mu)\frac{(4+\alpha)}{\beta}=2-\lambda-(2-\mu)\frac{4+\alpha}{\beta}=\frac{2\beta-\beta\lambda-8-2\alpha+\mu(4+\alpha)}{\beta} \tag{56}$$

for time:

$$-3+2\frac{4+\alpha}{\beta}=\frac{-3\beta+8+2\alpha}{\beta}=\frac{8-3\beta+2\alpha}{\beta} \tag{57}$$

Hence:

$$[B]=[M]^{\frac{\beta(2+\lambda)-\mu(4+\alpha)}{\beta}}[L]^{\frac{2\beta-\beta\lambda-8-2\alpha+\mu(4+\alpha)}{\beta}}[T]^{\frac{8-3\beta+2\alpha}{\beta}} \tag{58}$$

summary:

v	u	p	A	B	m	t
$M^{-1}L$	LT^{-1}	MLT^{-2}	$M^{\frac{\mu}{\beta}}L^{\frac{2-\mu}{\beta}}T^{-\frac{2}{\beta}-\tau}$	$M^{\frac{\beta(2+\lambda)-\mu(4+\alpha)}{\beta}}L^{\frac{2\beta-\beta\lambda-8-2\alpha+\mu(4+\alpha)}{\beta}}T^{\frac{8-3\beta+2\alpha}{\beta}}$	M	T

this is exactly SH doch, but they define the mass

$$M_{sh} = ML^2 \tag{59}$$

substituting ML^2 in their result will give our results for A, B.
we re-write for convenience:

v	u	p	A	B	m	t
$M^{-1}L$	LT^{-1}	MLT^{-2}	$M^{a_1}L^{a_2}T^{a_3}$	$M^{b_1}L^{b_2}T^{b_3}$	M	T

Hence, we have $N = 7$ quantities in the problem, with $K = 3$ units. Hence we have $N - K = 4$ dimensionless quantities.

$$\xi = mA^aB^bt^c \tag{60}$$

no M units:

$$1 + aa_1 + bb_1 = 0 \tag{61}$$

no L units:

$$aa_2 + bb_2 = 0 \tag{62}$$

from which

$$b = -\frac{a_2}{b_2}a \tag{63}$$

putting it in the first:

$$1 + aa_1 - \frac{a_2}{b_2}ab_1 = 0 \tag{64}$$

$$a\left(a_1 - \frac{a_2b_1}{b_2}\right) = -1 \tag{65}$$

$$a = \frac{-1}{a_1 - \frac{a_2b_1}{b_2}} \tag{66}$$

$$a = \frac{b_2}{a_2b_1 - a_1b_2} \tag{67}$$

$$b = -\frac{a_2}{b_2}a = -\frac{a_2}{b_2}\frac{b_2}{a_2b_1 - a_1b_2} \tag{68}$$

$$b = -\frac{a_2}{a_2b_1 - a_1b_2} \tag{69}$$

no T units:

$$aa_3 + bb_3 + c = 0 \tag{70}$$

$$c = -aa_3 - bb_3 \tag{71}$$

$$c = -\frac{b_2}{a_2 b_1 - a_1 b_2} a_3 + \frac{a_2}{a_2 b_1 - a_1 b_2} b_3 \tag{72}$$

$$c = \frac{a_2 b_3 - b_2 a_3}{a_2 b_1 - a_1 b_2} \tag{73}$$

using:

v	u	p	A	B	m	t
$M^{-1}L$	LT^{-1}	MLT^{-2}	$M^{\frac{\mu}{\beta}} L^{\frac{2-\mu}{\beta}} T^{-\frac{2}{\beta}-\tau}$	$M^{\frac{\beta(2+\lambda)-\mu(4+\alpha)}{\beta}} L^{\frac{2\beta-\beta\lambda-8-2\alpha+\mu(4+\alpha)}{\beta}} T^{\frac{8-3\beta+2\alpha}{\beta}}$	M	T

$$a_2 b_1 - a_1 b_2 = \left(\frac{2-\mu}{\beta}\right) \frac{\beta(2+\lambda) - \mu(4+\alpha)}{\beta} - \frac{\mu}{\beta} \frac{2\beta - \beta\lambda - 8 - 2\alpha + \mu(4+\alpha)}{\beta} \tag{74}$$

$$= \frac{\beta(4+2\lambda-2\mu-\mu\lambda) - \mu(8+2\alpha-4\mu-\alpha\mu) - 2\beta\mu + \beta\lambda\mu + 8\mu + 2\alpha\mu - 4\mu^2 - \mu^2\alpha}{\beta^2} \tag{75}$$

$$= \frac{\beta(4+2\lambda-2\mu-\mu\lambda) - 2\beta\mu + \beta\lambda\mu}{\beta^2} = \frac{4+2\lambda-4\mu}{\beta} \tag{76}$$

$$\boxed{a_2 b_1 - a_1 b_2 = \frac{4+2\lambda-4\mu}{\beta}} \tag{77}$$

$$a = \frac{b_2}{a_2 b_1 - a_1 b_2} = \frac{\beta}{4+2\lambda-4\mu} \frac{2\beta - \beta\lambda - 8 - 2\alpha + \mu(4+\alpha)}{\beta} = \frac{2\beta - \beta\lambda - 8 - 2\alpha + \mu(4+\alpha)}{4+2\lambda-4\mu} \tag{78}$$

$$b = -\frac{a_2}{a_2 b_1 - a_1 b_2} = -\frac{\beta}{4+2\lambda-4\mu} \frac{2-\mu}{\beta} = \frac{\mu-2}{4+2\lambda-4\mu} \tag{79}$$

$$
\begin{aligned}
c &= \frac{a_2 b_3 - b_2 a_3}{a_2 b_1 - a_1 b_2} && (80)\\
&= \frac{\beta}{4+2\lambda-4\mu}\left[\left(\frac{2-\mu}{\beta}\right)\left(\frac{8-3\beta+2\alpha}{\beta}\right)-\left(\frac{2\beta-\beta\lambda-8-2\alpha+\mu\left(4+\alpha\right)}{\beta}\right)\left(-\frac{2}{\beta}-\tau\right)\right] && (81)\\
&= \frac{1}{4+2\lambda-4\mu}\frac{1}{\beta}\left[\left(2-\mu\right)\left(8-3\beta+2\alpha\right)+\left(2+\tau\beta\right)\left(2\beta-\beta\lambda-8-2\alpha+\mu\left(4+\alpha\right)\right)\right] && (82)\\
&= \frac{1}{4+2\lambda-4\mu}\frac{1}{\beta}\left[16-6\beta+4\alpha-8\mu+3\beta\mu-2\mu\alpha+4\beta-2\beta\lambda-16-4\alpha+2\mu\left(4+\alpha\right)+..+\tau\beta\left(2\beta-\beta\lambda-8\right.\right. && (83)\\
&= \frac{1}{4+2\lambda-4\mu}\frac{1}{\beta}\left[-2\beta+3\beta\mu-2\beta\lambda+\tau\beta\left(2\beta-\beta\lambda-8-2\alpha+\mu\left(4+\alpha\right)\right)\right] && (84)\\
&= \frac{1}{4+2\lambda-4\mu}\left[-2+3\mu-2\lambda+\tau\left(2\beta-\beta\lambda-8-2\alpha+\mu\left(4+\alpha\right)\right)\right] && (85)\\
&= \frac{1}{4+2\lambda-4\mu}\left[3\mu-2\lambda-2+\tau\left(2\left(\beta-\alpha-4\right)-\beta\lambda+\mu\left(4+\alpha\right)\right)\right] && (86)
\end{aligned}
$$

hence:

$$\xi = mA^aB^bt^c \tag{87}$$

$$\boxed{\xi = m\left(A^{2\beta-\beta\lambda-8-2\alpha+\mu(4+\alpha)}B^{\mu-2}t^{3\mu-2\lambda-2+\tau(2(\beta-\alpha-4)-\beta\lambda+\mu(4+\alpha))}\right)^{\frac{1}{4+2\lambda-4\mu}} \quad (88)}$$

using:

$$V = vA^aB^bt^c \tag{89}$$

v	u	p	A	B	m	t
$M^{-1}L$	LT^{-1}	MLT^{-2}	$M^{a_1}L^{a_2}T^{a_3}$	$M^{b_1}L^{b_2}T^{b_3}$	M	T

no M units:

$$-1 + aa_1 + bb_1 = 0 \tag{90}$$

no L units:

$$1 + aa_2 + bb_2 = 0 \tag{91}$$

summing those:

$$a\left(a_1+a_2\right) + b\left(b_1+b_2\right) = 0 \tag{92}$$

$$b = -a\frac{a_1+a_2}{b_1+b_2} \tag{93}$$

plugging in the first:

$$-1+aa_1-a\frac{a_1+a_2}{b_1+b_2}b_1=0 \tag{94}$$

$$a\left(a_1-\frac{a_1+a_2}{b_1+b_2}b_1\right)=1 \tag{95}$$

$$a\left(\frac{a_1b_1+a_1b_2-a_1b_1-a_2b_1}{b_1+b_2}\right)=1 \tag{96}$$

$$a\left(\frac{a_1b_2-a_2b_1}{b_1+b_2}\right)=1 \tag{97}$$

$$a=\frac{b_1+b_2}{a_1b_2-a_2b_1} \tag{98}$$

$$a=-\frac{b_1+b_2}{a_2b_1-a_1b_2} \tag{99}$$

and hence:

$$b=-a\frac{a_1+a_2}{b_1+b_2} \tag{100}$$

$$b=\frac{b_1+b_2}{a_2b_1-a_1b_2}\frac{a_1+a_2}{b_1+b_2} \tag{101}$$

$$b=\frac{a_1+a_2}{a_2b_1-a_1b_2} \tag{102}$$

no T units:

$$aa_3+bb_3+c=0 \tag{103}$$

$$c=-aa_3-bb_3 \tag{104}$$

$$c=\frac{b_1+b_2}{a_2b_1-a_1b_2}a_3-\frac{a_1+a_2}{a_2b_1-a_1b_2}b_3 \tag{105}$$

$$c=\frac{a_3\left(b_1+b_2\right)-b_3\left(a_1+a_2\right)}{a_2b_1-a_1b_2} \tag{106}$$

we already saw:

$$a_2b_1-a_1b_2=\frac{4+2\lambda-4\mu}{\beta} \tag{107}$$

using:

v	u	p	A	B	m	t
$M^{-1}L$	LT^{-1}	MLT^{-2}	$M^{\frac{\mu}{\beta}}L^{\frac{2-\mu}{\beta}}T^{-\frac{2}{\beta}-\tau}$	$M^{\frac{\beta(2+\lambda)-\mu(4+\alpha)}{\beta}}L^{\frac{2\beta-\beta\lambda-8-2\alpha+\mu(4+\alpha)}{\beta}}T^{\frac{8-3\beta+2\alpha}{\beta}}$	M	T

we already saw

$$a_2 b_1 - a_1 b_2 = \frac{4 + 2\lambda - 4\mu}{\beta} \tag{108}$$

$$\begin{align}
a &= -\frac{b_1 + b_2}{a_1 b_2 - a_2 b_1} \tag{109} \\
&= -\frac{\beta}{4 + 2\lambda - 4\mu} \frac{\beta (2 + \lambda) - \mu (4 + \alpha) + 2\beta - \beta\lambda - 8 - 2\alpha + \mu (4 + \alpha)}{\beta} \tag{110} \\
&= -\frac{\beta (2 + \lambda) - \mu (4 + \alpha) + 2\beta - \beta\lambda - 8 - 2\alpha + \mu (4 + \alpha)}{4 + 2\lambda - 4\mu} \tag{111} \\
&= -\frac{4\beta - 8 - 2\alpha}{4 + 2\lambda - 4\mu} = \frac{4 + \alpha - 2\beta}{2 + \lambda - 2\mu} \tag{112}
\end{align}$$

$$\begin{align}
b &= \frac{a_1 + a_2}{a_2 b_1 - a_1 b_2} \tag{113} \\
&= \frac{\beta}{4 + 2\lambda - 4\mu} \frac{\mu + 2 - \mu}{\beta} = \frac{1}{2 + \lambda - 2\mu} \tag{114}
\end{align}$$

$$\begin{align}
c &= -a a_3 - b b_3 \tag{115} \\
&= -\frac{1}{2 + \lambda - 2\mu} \left((4 + \alpha - 2\beta) a_3 + b_3 \right) \tag{116} \\
&= -\frac{1}{2 + \lambda - 2\mu} \left((4 + \alpha - 2\beta) \left(-\frac{2}{\beta} - \tau \right) + \frac{8 - 3\beta + 2\alpha}{\beta} \right) \tag{117} \\
&= \frac{1}{2 + \lambda - 2\mu} \left((4 + \alpha - 2\beta) \left(\frac{2}{\beta} + \tau \right) - \frac{8 - 3\beta + 2\alpha}{\beta} \right) \tag{118} \\
&= \frac{1}{2 + \lambda - 2\mu} \left(\tau (4 + \alpha - 2\beta) + 2 \left(\frac{4 + \alpha - 2\beta}{\beta} \right) + \frac{3\beta - 8 - 2\alpha}{\beta} \right) \tag{119} \\
&= \frac{1}{2 + \lambda - 2\mu} \left(\tau (4 + \alpha - 2\beta) + \frac{8 + 2\alpha - 4\beta + 3\beta - 8 - 2\alpha}{\beta} \right) \tag{120} \\
&= \frac{1}{2 + \lambda - 2\mu} \left(\tau (4 + \alpha - 2\beta) - 1 \right) \tag{121}
\end{align}$$

hence:

$$V = v A^a B^b t^c \tag{122}$$

$$\boxed{V = v \left(A^{4 + \alpha - 2\beta} B t^{\tau (4 + \alpha - 2\beta) - 1} \right)^{\frac{1}{2 + \lambda - 2\mu}}} \tag{123}$$

using:

$$U = u A^a B^b t^c \tag{124}$$

v	u	p	A	B	m	t
$M^{-1}L$	LT^{-1}	MLT^{-2}	$M^{a_1}L^{a_2}T^{a_3}$	$M^{b_1}L^{b_2}T^{b_3}$	M	T

no M units:

$$aa_1 + bb_1 = 0 \tag{125}$$

no L units:

$$1 + aa_2 + bb_2 = 0 \tag{126}$$

the first gives

$$b = -a\frac{a_1}{b_1} \tag{127}$$

put in the second

$$1 + aa_2 - a\frac{a_1}{b_1}b_2 = 0 \tag{128}$$

$$a\left(a_2 - \frac{a_1}{b_1}b_2\right) = -1 \tag{129}$$

$$a = \frac{-1}{a_2 - \frac{a_1}{b_1}b_2} \tag{130}$$

$$a = -\frac{b_1}{a_2b_1 - a_1b_2} \tag{131}$$

hence:

$$b = -a\frac{a_1}{b_1} \tag{132}$$

$$b = \frac{b_1}{a_2b_1 - a_1b_2}\frac{a_1}{b_1} \tag{133}$$

$$b = \frac{a_1}{a_2b_1 - a_1b_2} \tag{134}$$

no T units:

$$-1 + aa_3 + bb_3 + c = 0 \tag{135}$$

$$c = 1 - aa_3 - bb_3 \tag{136}$$

using:

v	u	p	A	B	m	t
$M^{-1}L$	LT^{-1}	MLT^{-2}	$M^{\frac{\mu}{\beta}}L^{\frac{2-\mu}{\beta}}T^{-\frac{2}{\beta}-\tau}$	$M^{\frac{\beta(2+\lambda)-\mu(4+\alpha)}{\beta}}L^{\frac{2\beta-\beta\lambda-8-2\alpha+\mu(4+\alpha)}{\beta}}T^{\frac{8-3\beta+2\alpha}{\beta}}$	M	T

we already saw

$$a_2b_1 - a_1b_2 = \frac{4 + 2\lambda - 4\mu}{\beta} \tag{137}$$

$$a = -\frac{b_1}{a_2 b_1 - a_1 b_2} \tag{138}$$

$$= -\frac{\beta}{4+2\lambda-4\mu}\frac{\beta(2+\lambda)-\mu(4+\alpha)}{\beta} \tag{139}$$

$$= -\frac{\beta(2+\lambda)-\mu(4+\alpha)}{4+2\lambda-4\mu} \tag{140}$$

$$= \frac{\mu(4+\alpha)-\beta(2+\lambda)}{4+2\lambda-4\mu} \tag{141}$$

$$b = \frac{a_1}{a_2 b_1 - a_1 b_2} \tag{142}$$

$$= \frac{\beta}{4+2\lambda-4\mu}\frac{\mu}{\beta} = \frac{\mu}{4+2\lambda-4\mu} \tag{143}$$

$$c = 1 - aa_3 - bb_3 \tag{144}$$

$$= 1 - \frac{1}{4+2\lambda-4\mu}\left[(\mu(4+\alpha)-\beta(2+\lambda))\left(-\frac{2}{\beta}-\tau\right)+\mu\left(\frac{8-3\beta+2\alpha}{\beta}\right)\right] \tag{145}$$

$$= 1 + \frac{1}{4+2\lambda-4\mu}\left[(\mu(4+\alpha)-\beta(2+\lambda))\left(\frac{2}{\beta}+\tau\right)+\mu\left(\frac{3\beta-8-2\alpha}{\beta}\right)\right] \tag{146}$$

$$= 1 + \frac{1}{4+2\lambda-4\mu}\left[\tau(\mu(4+\alpha)-\beta(2+\lambda))+(\mu(4+\alpha)-\beta(2+\lambda))\frac{2}{\beta}+\mu\left(\frac{3\beta-8-2\alpha}{\beta}\right)\right] \tag{147}$$

$$= 1 + \frac{1}{4+2\lambda-4\mu}\left[\tau(\mu(4+\alpha)-\beta(2+\lambda))+\frac{8\mu+2\mu\alpha-4\beta-2\lambda\beta+3\mu\beta-8\mu-2\mu\alpha}{\beta}\right] \tag{148}$$

$$= 1 + \frac{1}{4+2\lambda-4\mu}\left[\tau(\mu(4+\alpha)-\beta(2+\lambda))+\frac{-4\beta-2\lambda\beta+3\mu\beta}{\beta}\right] \tag{149}$$

$$= 1 + \frac{\tau(\mu(4+\alpha)-\beta(2+\lambda))}{4+2\lambda-4\mu} + \frac{-4-2\lambda+3\mu}{4+2\lambda-4\mu} \tag{150}$$

$$= 1 + \frac{\tau(\mu(4+\alpha)-\beta(2+\lambda))}{4+2\lambda-4\mu} + \frac{-4-2\lambda+4\mu-\mu}{4+2\lambda-4\mu} \tag{151}$$

$$= 1 + \frac{\tau(\mu(4+\alpha)-\beta(2+\lambda))-\mu}{4+2\lambda-4\mu} - 1 \tag{152}$$

$$= \frac{\tau(\mu(4+\alpha)-\beta(2+\lambda))-\mu}{4+2\lambda-4\mu} \tag{153}$$

hence:

$$U = uA^a B^b t^c \tag{154}$$

$$\boxed{U = u\left(A^{\mu(4+\alpha)-\beta(2+\lambda)}B^{\mu}t^{\tau(\mu(4+\alpha)-\beta(2+\lambda))-\mu}\right)^{\frac{1}{4+2\lambda-4\mu}}} \tag{155}$$

using:

$$P = pA^a B^b t^c \tag{156}$$

v	u	p	A	B	m	t
$M^{-1}L$	LT^{-1}	MLT^{-2}	$M^{a_1}L^{a_2}T^{a_3}$	$M^{b_1}L^{b_2}T^{b_3}$	M	T

no M units:

$$1 + aa_1 + bb_1 = 0 \tag{157}$$

no L units:

$$1 + aa_2 + bb_2 = 0 \tag{158}$$

subbing the two:

$$a\left(a_2 - a_1\right) + b\left(b_2 - b_1\right) = 0 \tag{159}$$

$$b = -a\frac{a_2 - a_1}{b_2 - b_1} \tag{160}$$

put it in the first:

$$1 + aa_1 - a\frac{a_2 - a_1}{b_2 - b_1}b_1 = 0 \tag{161}$$

$$a\left(a_1 - \frac{a_2 - a_1}{b_2 - b_1}b_1\right) = -1 \tag{162}$$

$$a\left(\frac{a_1b_2 - a_1b_1 - a_2b_1 + a_1b_1}{b_2 - b_1}\right) = -1 \tag{163}$$

$$a\left(\frac{a_1b_2 - a_2b_1}{b_2 - b_1}\right) = -1 \tag{164}$$

$$a = \frac{b_2 - b_1}{a_2b_1 - a_1b_2} \tag{165}$$

$$b = -a\frac{a_2 - a_1}{b_2 - b_1} \tag{166}$$

$$b = -\frac{b_2 - b_1}{a_2b_1 - a_1b_2}\frac{a_2 - a_1}{b_2 - b_1} \tag{167}$$

$$b = \frac{a_1 - a_2}{a_2b_1 - a_1b_2} \tag{168}$$

no T units:

$$-2 + aa_3 + bb_3 + c = 0 \tag{169}$$

$$c = 2 - aa_3 - bb_3 \tag{170}$$

using:

v	u	p	A	B	m	t
$M^{-1}L$	LT^{-1}	MLT^{-2}	$M^{\frac{\mu}{\beta}}L^{\frac{2-\mu}{\beta}}T^{-\frac{2}{\beta}-\tau}$	$M^{\frac{\beta(2+\lambda)-\mu(4+\alpha)}{\beta}}L^{\frac{2\beta-\beta\lambda-8-2\alpha+\mu(4+\alpha)}{\beta}}T^{\frac{8-3\beta+2\alpha}{\beta}}$	M	T

we already saw

$$a_2 b_1 - a_1 b_2 = \frac{4 + 2\lambda - 4\mu}{\beta} \tag{171}$$

$$\begin{aligned}
a &= \frac{b_2 - b_1}{a_2 b_1 - a_1 b_2} && (172)\\
&= \frac{\beta}{4 + 2\lambda - 4\mu} \frac{2\beta - \beta\lambda - 8 - 2\alpha + \mu\,(4 + \alpha) - \beta\,(2 + \lambda) + \mu\,(4 + \alpha)}{\beta} && (173)\\
&= \frac{2\beta - \beta\lambda - 8 - 2\alpha + \mu\,(4 + \alpha) - 2\beta - \beta\lambda + \mu\,(4 + \alpha)}{4 + 2\lambda - 4\mu} && (174)\\
&= \frac{-2\beta\lambda - 8 - 2\alpha + 2\mu\,(4 + \alpha)}{4 + 2\lambda - 4\mu} = \frac{\mu\,(4 + \alpha) - \beta\lambda - 4 - \alpha}{2 + \lambda - 2\mu} && (175)
\end{aligned}$$

$$\begin{aligned}
b &= \frac{a_1 - a_2}{a_2 b_1 - a_1 b_2} && (176)\\
&= \frac{\beta}{4 + 2\lambda - 4\mu} \frac{\mu - (2 - \mu)}{\beta} && (177)\\
&= \frac{2\mu - 2}{4 + 2\lambda - 4\mu} && (178)\\
&= \frac{\mu - 1}{2 + \lambda - 2\mu} && (179)
\end{aligned}$$

$$c = 2 - aa_3 - bb_3 \tag{180}$$

$$= 2 - \frac{1}{2+\lambda-2\mu}\left[\left(\mu\left(4+\alpha\right)-\beta\lambda-4-\alpha\right)\left(-\frac{2}{\beta}-\tau\right)+\left(\mu-1\right)\left(\frac{8-3\beta+2\alpha}{\beta}\right)\right] \tag{181}$$

$$= 2 + \frac{1}{2+\lambda-2\mu}\left[\left(\mu\left(4+\alpha\right)-\beta\lambda-4-\alpha\right)\left(\frac{2}{\beta}+\tau\right)+\left(1-\mu\right)\left(\frac{8-3\beta+2\alpha}{\beta}\right)\right] \tag{182}$$

$$= 2 + \frac{1}{2+\lambda-2\mu}\left[\tau\left(\mu\left(4+\alpha\right)-\beta\lambda-4-\alpha\right)+\frac{2\left(\mu\left(4+\alpha\right)-\beta\lambda-4-\alpha\right)+\left(1-\mu\right)\left(8-3\beta+2\alpha\right)}{\beta}\right] \tag{183}$$

$$= 2 + \frac{1}{2+\lambda-2\mu}\left[\tau\left(\mu\left(4+\alpha\right)-\beta\lambda-4-\alpha\right)+\frac{8\mu+2\mu\alpha-2\beta\lambda-8-2\alpha+8-3\beta+2\alpha-8\mu+3\beta\mu-2\mu}{\beta}\right. \tag{184}$$

$$= 2 + \frac{1}{2+\lambda-2\mu}\left[\tau\left(\mu\left(4+\alpha\right)-\beta\lambda-4-\alpha\right)+\frac{-2\beta\lambda-3\beta+3\beta\mu}{\beta}\right] \tag{185}$$

$$= 2 + \frac{1}{2+\lambda-2\mu}\left[\tau\left(\mu\left(4+\alpha\right)-\beta\lambda-4-\alpha\right)-2\lambda-3+3\mu\right] \tag{186}$$

$$= \cancel{2} + \frac{1}{2+\lambda-2\mu}\left[\tau\left(\mu\left(4+\alpha\right)-\beta\lambda-4-\alpha\right)\cancel{-2\lambda-4+4\mu}+1-\mu\right] \tag{187}$$

$$= \frac{1}{2+\lambda-2\mu}\left[\tau\left(\mu\left(4+\alpha\right)-\beta\lambda-4-\alpha\right)+1-\mu\right] \tag{188}$$

hence:

$$P = pA^aB^bt^c \tag{189}$$

$$\boxed{P = p\left(A^{\mu(4+\alpha)-\beta\lambda-4-\alpha}B^{\mu-1}t^{\tau(\mu(4+\alpha)-\beta\lambda-4-\alpha)+1-\mu}\right)^{\frac{1}{2+\lambda-2\mu}} \quad (190)}$$

summary

$$\xi = m\left(A^{2\beta-\beta\lambda-8-2\alpha+\mu(4+\alpha)}B^{\mu-2}t^{3\mu-2\lambda-2+\tau(2(\beta-\alpha-4)-\beta\lambda+\mu(4+\alpha))}\right)^{\frac{1}{4+2\lambda-4\mu}} \tag{191}$$

$$V = v\left(A^{4+\alpha-2\beta}Bt^{\tau(4+\alpha-2\beta)-1}\right)^{\frac{1}{2+\lambda-2\mu}} \tag{192}$$

$$U = u\left(A^{\mu(4+\alpha)-\beta(2+\lambda)}B^{\mu}t^{\tau(\mu(4+\alpha)-\beta(2+\lambda))-\mu}\right)^{\frac{1}{4+2\lambda-4\mu}} \tag{193}$$

$$P = p\left(A^{\mu(4+\alpha)-\beta\lambda-4-\alpha}B^{\mu-1}t^{\tau(\mu(4+\alpha)-\beta\lambda-4-\alpha)+1-\mu}\right)^{\frac{1}{2+\lambda-2\mu}} \tag{194}$$

which we re-write as:

$$\xi = mA^aB^bt^c \tag{195}$$

$$v = VA^{a_1}B^{b_1}t^{c_1} \tag{196}$$

$$u = UA^{a_2}B^{b_2}t^{c_2} \tag{197}$$

$$p = PA^{a_3}B^{b_3}t^{c_3} \tag{198}$$

1.2 Derivation of self similar ODE

$$\xi = mA^aB^bt^c \tag{199}$$

$$\frac{\partial \xi}{\partial m} = \frac{\xi}{m} = A^aB^bt^c \tag{200}$$

$$\frac{\partial}{\partial m}\left(\frac{\xi}{m}\right) = 0 \tag{201}$$

$$\frac{\partial \xi}{\partial t} = c\frac{\xi}{t} \tag{202}$$

The continuity PDE is:

$$\frac{\partial v}{\partial t} - \frac{\partial u}{\partial m} = 0 \tag{203}$$

now:

$$\frac{\partial v}{\partial t} = \frac{\partial}{\partial t}\left(VA^{a_1}B^{b_1}t^{c_1}\right) = A^{a_1}B^{b_1}t^{c_1}\left(V'\frac{\partial \xi}{\partial t} + \frac{V}{t}c_1\right) \tag{204}$$

$$= A^{a_1}B^{b_1}t^{c_1}\left(c\frac{\xi}{t}V' + \frac{V}{t}c_1\right) \tag{205}$$

$$= A^{a_1}B^{b_1}t^{c_1-1}\left(c\xi V' + c_1V\right) \tag{206}$$

now:

$$\frac{\partial u}{\partial m} = \frac{\partial}{\partial m}\left(UA^{a_2}B^{b_2}t^{c_2}\right) \tag{207}$$

$$= A^{a_2}B^{b_2}t^{c_2}\frac{\partial \xi}{\partial m}U' \tag{208}$$

$$= A^{a_2}B^{b_2}t^{c_2}\left(A^aB^bt^c\right)U' \tag{209}$$

$$= A^{a_2+a}B^{b_2+b}t^{c_2+c}U' \tag{210}$$

$$\frac{\partial v}{\partial t} - \frac{\partial u}{\partial m} = 0 \tag{211}$$

$$A^{a_1}B^{b_1}t^{c_1-1}\left(c\xi V' + c_1 V\right) - A^{a_2+a}B^{b_2+b}t^{c_2+c}U' = 0 \tag{212}$$

all units cancel:

$$a_2 + a = \frac{-\mu\left(4+\alpha\right) + \beta\left(2+\lambda\right) + 2\beta - \beta\lambda - 8 - 2\alpha + \mu\left(4+\alpha\right)}{4 + 2\lambda - 4\mu} \tag{213}$$

$$= \frac{4\beta - 8 - 2\alpha}{4 + 2\lambda - 4\mu} \tag{214}$$

$$= \frac{2\beta - 4 - \alpha}{2 + \lambda - 2\mu} \tag{215}$$

$$= a_1 \tag{216}$$

similarly, we must prove that:

$$a_2 = a_1 - a,\ b_2 = b_1 - b,\ c_2 = c_1 - c - 1 \tag{217}$$

$$a_2 = a_1 - a,\ b_2 = b_1 - b,\ c_2 = c_1 - c - 1 \tag{218}$$

$$c\xi V' + c_1 V - U' = 0 \tag{219}$$

momentum equation:

$$\frac{\partial u}{\partial t} = -\frac{\partial p}{\partial m} \tag{220}$$

derivatives exactly as done for v above:

$$\frac{\partial u}{\partial t} = A^{a_2}B^{b_2}t^{c_2-1}\left(c\xi U' + c_2 U\right) \tag{221}$$

$$\frac{\partial p}{\partial m} = A^{a_3+a}B^{b_3+b}t^{c_3+c}P' \tag{222}$$

$$\frac{\partial u}{\partial t} + \frac{\partial p}{\partial m} = 0 \tag{223}$$

$$A^{a_2}B^{b_2}t^{c_2-1}\left(c\xi U' + c_2 U\right) + A^{a_3+a}B^{b_3+b}t^{c_3+c}P' = 0 \tag{224}$$

we must prove that:

$$a_3 = a_2 - a,\ b_3 = b_2 - b,\ c_3 = c_2 - c - 1 \tag{225}$$

using the previous relations from the mass equation:

$$a_3 = a_1 - 2a,\ b_3 = b_1 - 2b,\ c_3 = c_1 - 2c - 2 \tag{226}$$

$$\boxed{\begin{gathered} a_3 = a_1 - 2a,\ b_3 = b_1 - 2b,\ c_3 = c_1 - 2c - 2 \qquad (227) \\ c\xi U' + c_2 U + P' = 0 \qquad (228) \end{gathered}}$$

energy equation:

$$\frac{\partial}{\partial t}\left(\frac{pv}{r}\right) + p\frac{\partial v}{\partial t} = \left(\frac{\beta}{4+\alpha}\right) B \frac{\partial}{\partial m}\left(v^{\lambda}\frac{\partial}{\partial m}\left(pv^{1-\mu}\right)^{\frac{4+\alpha}{\beta}}\right) \tag{229}$$

$$\frac{\partial}{\partial t}\left(\frac{pv}{r}\right) + p\frac{\partial v}{\partial t} = \frac{v}{r}\frac{\partial p}{\partial t} + \frac{p}{r}\frac{\partial v}{\partial t} + p\frac{\partial v}{\partial t} = \left(1+\frac{1}{r}\right)p\frac{\partial v}{\partial t} + \frac{v}{r}\frac{\partial p}{\partial t} \tag{230}$$

hence:

$$\left(1+\frac{1}{r}\right)p\frac{\partial v}{\partial t} + \frac{v}{r}\frac{\partial p}{\partial t} = \left(\frac{\beta}{4+\alpha}\right) B \frac{\partial}{\partial m}\left(v^{\lambda}\frac{\partial}{\partial m}\left(pv^{1-\mu}\right)^{\frac{4+\alpha}{\beta}}\right) \tag{231}$$

LHS:
using

$$\frac{\partial v}{\partial t} = A^{a_1}B^{b_1}t^{c_1-1}\left(c\xi V' + c_1 V\right) \tag{232}$$

$$\frac{\partial p}{\partial t} = A^{a_3}B^{b_3}t^{c_3-1}\left(c\xi P' + c_3 P\right) \tag{233}$$

$$\begin{aligned} \left(1+\frac{1}{r}\right)p\frac{\partial v}{\partial t} + \frac{v}{r}\frac{\partial p}{\partial t} &= \left(1+\frac{1}{r}\right)PA^{a_3}B^{b_3}t^{c_3}A^{a_1}B^{b_1}t^{c_1-1}\left(c\xi V' + c_1 V\right) && (234) \\ &+ \frac{1}{r}VA^{a_1}B^{b_1}t^{c_1}A^{a_3}B^{b_3}t^{c_3-1}\left(c\xi P' + c_3 P\right) && (235) \\ &= A^{a_1+a_3}B^{b_1+b_3}t^{c_1+c_3-1}\left[c\xi\left(\left(1+\frac{1}{r}\right)PV' + \frac{1}{r}VP'\right) + PV\left(\left(1+\frac{1}{r}\right)c_1 + \frac{c_3}{r}\right)\right] && (236) \end{aligned}$$

perfect LHS of S.H. eq. 30c.
RHS:

$$\frac{\text{rhs}}{B} = \left(\frac{\beta}{4+\alpha}\right)\frac{\partial}{\partial m}\left(v^{\lambda}\frac{\partial}{\partial m}\left(pv^{1-\mu}\right)^{\frac{4+\alpha}{\beta}}\right) \tag{237}$$

$$=\frac{1}{n}\frac{\partial}{\partial m}\left(v^{\lambda}\frac{\partial}{\partial m}\left(pv^{k}\right)^{n}\right) = \frac{1}{n}n\frac{\partial}{\partial m}\left(v^{\lambda}\left(pv^{k}\right)^{n-1}\frac{\partial}{\partial m}\left(pv^{k}\right)\right) \tag{238}$$

$$= \frac{\partial}{\partial m}\left(v^{\lambda}\left(pv^{k}\right)^{n-1}\left(v^{k}p' + kpv^{k-1}v'\right)\right) \tag{239}$$

$$= \frac{\partial}{\partial m}\left(p^{n-1}v^{k(n-1)+\lambda+k-1}\left(vp' + kpv'\right)\right) \tag{240}$$

$$= \frac{\partial}{\partial m}\left(p^{n-1}v^{q}\left(vp' + kpv'\right)\right) \tag{241}$$

$$= \left(vp' + kpv'\right)\frac{\partial}{\partial m}\left(p^{n-1}v^{q}\right) \tag{242}$$

$$+ p^{n-1}v^{q}\frac{\partial}{\partial m}\left(vp' + kpv'\right) \tag{243}$$

$$= \left(vp' + kpv'\right)\left(p^{n-1}qv^{q-1}v' + (n-1)\,p^{n-2}v^{q}p'\right) \tag{244}$$

$$+ p^{n-1}v^{q}\left[v'p' + vp'' + kp'v' + kpv''\right] \tag{245}$$

$$= v^{q-1}p^{n-2}\left(vp' + kpv'\right)\left(qpv' + (n-1)\,vp'\right) \tag{246}$$

$$+ p^{n-1}v^{q}\left[(k+1)\,v'p' + vp'' + kpv''\right] \tag{247}$$

$$\frac{\text{rhs}}{B} = v^{q-1}p^{n-2}\left(vp' + kpv'\right)\left(qpv' + (n-1)\,vp'\right) \tag{248}$$

$$+ p^{n-1}v^{q}\left[(k+1)\,v'p' + vp'' + kpv''\right] \tag{249}$$

$$= v^{q-1}p^{n-2}\left[\left(vp' + kpv'\right)\left(qpv' + (n-1)\,vp'\right) + pv\left((k+1)\,v'p' + vp'' + kpv''\right)\right] \tag{250}$$

$$= v^{q-1}p^{n-2}\left[qpvv'p' + (n-1)\,v^{2}p'^{2} + kqp^{2}v'^{2} + k\,(n-1)\,pvv'p' + pv\left((k+1)\,v'p' + vp'' + kpv''\right)\right] \tag{251}$$

$$= v^{q-1}p^{n-2}\left[pv\left(vp'' + kpv''\right) + kqp^{2}v'^{2} + (n-1)\,v^{2}p'^{2} + (q + k\,(n-1) + k + 1)\,pvp'v'\right] \tag{252}$$

$$= v^{q-1}p^{n-2}\left[pv\left(vp'' + kpv''\right) + kqp^{2}v'^{2} + (n-1)\,v^{2}p'^{2} + (q + kn + 1)\,pvp'v'\right] \tag{253}$$

$$= v^{q}p^{n-1}\left(kpv'' + vp''\right) \tag{254}$$

$$+ kqv^{q-1}p^{n}v'^{2} \tag{255}$$

$$+ (n-1)\,v^{q+1}p^{n-2}p'^{2} \tag{256}$$

$$+ (q + kn + 1)\,v^{q}p^{n-1}p'v' \tag{257}$$

$$n = \frac{4+\alpha}{\beta} \tag{258}$$

$$n - 1 = \frac{4+\alpha-\beta}{\beta} \tag{259}$$

$$n-2=\frac{4+\alpha-2\beta}{\beta} \tag{260}$$

$$k=1-\mu \tag{261}$$

$$q=k\left(n-1\right)+\lambda+k-1 \tag{262}$$

$$q=kn+\lambda-1 \tag{263}$$

$$q+kn+1=2nk+\lambda \tag{264}$$

using

$$\frac{\partial v}{\partial m}=A^{a_1+a}B^{b_1+b}t^{c_1+c}V' \tag{265}$$

$$\frac{\partial p}{\partial m}=A^{a_3+a}B^{b_3+b}t^{c_3+c}P' \tag{266}$$

plugging it and ASSUMING all unit factors will cancel, the equation is will read:

$$c\xi\left[\left(1+\frac{1}{r}\right)PV'+\frac{1}{r}VP'\right]+PV\left[\left(1+\frac{1}{r}\right)c_1+\frac{c_3}{r}\right] \tag{267}$$

$$=V^qP^{n-1}\left(kPV''+VP''\right)+kqV^{q-1}P^nV'^2+\left(n-1\right)V^{q+1}P^{n-2}P'^2+\left(q+kn+1\right)V^qP^{n-1}P'V' \tag{268}$$

$$V^qP^{n-1}\left(kPV''+VP''\right)=\left(c\xi\left[\left(1+\frac{1}{r}\right)PV'+\frac{1}{r}VP'\right]+PV\left[\left(1+\frac{1}{r}\right)c_1+\frac{c_3}{r}\right]\right) \tag{269}$$

$$-kqV^{q-1}P^nV'^2-\left(n-1\right)V^{q+1}P^{n-2}P'^2-\left(q+kn+1\right)V^qP^{n-1}P'V' \tag{270}$$

$$kPV''+VP''=V^{-q}P^{1-n}\left(c\xi\left[\left(1+\frac{1}{r}\right)PV'+\frac{1}{r}VP'\right]+PV\left[\left(1+\frac{1}{r}\right)c_1+\frac{c_3}{r}\right]\right) \tag{271}$$

$$-kqV^{-1}PV'^2-\left(n-1\right)VP^{-1}P'^2-\left(q+kn+1\right)P'V' \tag{272}$$

In order to find a connection between P'' and V'', we notice the mass equation:

$$c\xi V'+c_1V-U'=0 \tag{273}$$

$$U'=c\xi V'+c_1V \tag{274}$$

$$U'' = c\xi V'' + (c_1 + c) V' \tag{275}$$

momentum equation:

$$c\xi U' + c_2 U + P' = 0 \tag{276}$$

$$P' = -c\xi U' - c_2 U \tag{277}$$

$$P' = -c\xi U' - c_2 U \tag{278}$$

$$P'' = -c\xi U'' - (c_2 + c) U' \tag{279}$$

$$P'' = -\, c\xi \left(c\xi V'' + (c_1 + c) V'\right) - (c_2 + c) U' \tag{280}$$

$$P'' = -\, c^2\xi^2 V'' - c\,(c_1 + c)\,\xi V' - (c_2 + c) U' \tag{281}$$

putting this in the energy equation:

$$kPV'' + VP'' = V^{-q}P^{1-n}\left(c\xi\left[\left(1+\frac{1}{r}\right)PV' + \frac{1}{r}VP'\right] + PV\left[\left(1+\frac{1}{r}\right)c_1 + \frac{c_3}{r}\right]\right) \tag{282}$$
$$-\, kqV^{-1}PV'^2 - (n-1)\,VP^{-1}P'^2 - (q + kn + 1)\,P'V' \tag{283}$$

$$kPV'' + VP'' = kPV'' - V\left[c^2\xi^2 V'' + c\,(c_1 + c)\,\xi V' + (c_2 + c)\,U'\right] \tag{284}$$
$$= V^{-q}P^{1-n}\left[c\xi\left(\left(1+\frac{1}{r}\right)PV' + \frac{1}{r}VP'\right) + PV\left(\left(1+\frac{1}{r}\right)c_1 + \frac{c_3}{r}\right)\right] \tag{285}$$
$$-\, kqV^{-1}PV'^2 - (n-1)\,VP^{-1}P'^2 - (q + kn + 1)\,P'V' \tag{286}$$

$$\left(kP - c^2\xi^2 V\right) V'' \tag{287}$$
$$= V^{-q}P^{1-n}\left[c\xi\left(\left(1+\frac{1}{r}\right)PV' + \frac{1}{r}VP'\right) + PV\left(\left(1+\frac{1}{r}\right)c_1 + \frac{c_3}{r}\right)\right] \tag{288}$$
$$-\, kqV^{-1}PV'^2 - (n-1)\,VP^{-1}P'^2 - (q + kn + 1)\,P'V' \tag{289}$$
$$+\, c\,(c_1 + c)\,\xi VV' + (c_2 + c)\,VU' \tag{290}$$

summary:

$$\frac{dV}{d\xi} = V' \tag{291}$$

$$\frac{dP}{d\xi} = P' \tag{292}$$

$$\frac{dU}{d\xi} = c\xi V' + c_1 V \tag{293}$$

$$\left(kP - c^2\xi^2 V\right)\frac{d^2V}{d\xi^2} = c\left(c_1 + c\right)\xi V V' + \left(c_2 + c\right) V \frac{dU}{d\xi} \tag{294}$$

$$- kqV^{-1}PV'^2 - (n-1)\, VP^{-1}P'^2 - (q + kn + 1)\, P'V' \tag{295}$$

$$+ V^{-q}P^{1-n}\left[c\xi\left(\left(1+\frac{1}{r}\right)PV' + \frac{1}{r}VP'\right) + PV\left(\left(1+\frac{1}{r}\right)c_1 + \frac{c_3}{r}\right)\right] \tag{296}$$

$$\frac{d^2P}{d\xi^2} = -c^2\xi^2\frac{d^2V}{d\xi^2} - c\left(c_1 + c\right)\xi V' - \left(c_2 + c\right)\frac{dU}{d\xi} \tag{297}$$

we have 5 variables (V, V', P, P', U) and 5 equations for their derivatives $\left(\frac{dV}{d\xi}, \frac{d^2V}{d\xi^2}, \frac{dP}{d\xi}, \frac{d^2P}{d\xi^2}, \frac{dU}{d\xi}\right)$ which are written in terms of the variables (V, V', P, P', U).

$$n = \frac{4+\alpha}{\beta} \tag{298}$$

$$n - 1 = \frac{4+\alpha-\beta}{\beta} \tag{299}$$

$$k = 1 - \mu \tag{300}$$

$$q = k\,(n-1) + \lambda + k - 1 \tag{301}$$

1.3 working with $\ln \xi$

$$\frac{df}{d\ln x} = \frac{df}{dx}\frac{dx}{d\ln x} = x\frac{df}{dx} \tag{302}$$

$$\frac{d\ln f}{d\ln x} = x\frac{d\ln f}{dx} = \frac{x}{f}\frac{df}{dx} \tag{303}$$

$$\frac{d\ln f}{d\ln x} = \frac{1}{f}\frac{df}{d\ln x} \tag{304}$$

hence:

$$\frac{df}{dx} = \frac{1}{x}\frac{df}{d\ln x} \tag{305}$$

$$\frac{df}{dx} = \frac{f}{x}\frac{d\ln f}{d\ln x} \tag{306}$$

now

$$\frac{d^2 f}{dx^2} = \frac{d}{dx}\left(\frac{1}{x}\frac{df}{d\ln x}\right) = -\frac{1}{x^2}\frac{df}{d\ln x} + \frac{1}{x}\frac{d}{dx}\left(\frac{df}{d\ln x}\right) \tag{307}$$

$$= -\frac{1}{x^2}\frac{df}{d\ln x} + \frac{1}{x}\left(\frac{1}{x}\frac{d}{d\ln x}\left(\frac{df}{d\ln x}\right)\right) \tag{308}$$

$$= \frac{1}{x^2}\left(\frac{d^2 f}{d\,(\ln x)^{\,2}} - \frac{df}{d\ln x}\right) \tag{309}$$

so we have the identities:

$$\frac{df}{dx} = \frac{1}{x}\frac{df}{d\ln x} \tag{310}$$

$$\frac{d^2 f}{dx^2} = \frac{1}{x^2}\left(\frac{d^2 f}{d\,(\ln x)^{\,2}} - \frac{df}{d\ln x}\right) \tag{311}$$

define

$$\eta = \ln\xi \tag{312}$$

eq are:

$$\frac{dU}{d\xi} = c\xi V' + c_1 V \tag{313}$$

which is

$$\frac{1}{\xi}\frac{dU}{d\eta} = c\frac{dV}{d\eta} + c_1 V \tag{314}$$

$$\frac{dU}{d\eta} = \xi\left(c\frac{dV}{d\eta} + c_1 V\right) \tag{315}$$

energy equation:

$$\left(kP - c^2\xi^2 V\right)\frac{d^2 V}{d\xi^2} = c\,(c_1 + c)\,\xi V V' + (c_2 + c)\,V\frac{dU}{d\xi} \tag{316}$$

$$-\,kqV^{-1}PV'^2 - (n-1)\,VP^{-1}P'^2 - (q + kn + 1)\,P'V' \tag{317}$$

$$+\,V^{-q}P^{1-n}\left[c\xi\left(\left(1 + \frac{1}{r}\right)PV' + \frac{1}{r}VP'\right) + PV\left(\left(1 + \frac{1}{r}\right)c_1 + \frac{c_3}{r}\right)\right] \tag{318}$$

$$\left(kP - c^2\xi^2 V\right)\frac{1}{\xi^2}\left(\frac{d^2 V}{d\eta^2} - \frac{dV}{d\eta}\right) = c\,(c_1 + c)\,V\frac{dV}{d\eta} + (c_2 + c)\,V\frac{dU}{d\xi} \tag{319}$$

$$-\,kqV^{-1}PV'^2 - (n-1)\,VP^{-1}P'^2 - (q + kn + 1)\,P'V' \tag{320}$$

$$+\,V^{-q}P^{1-n}\left[c\left(\left(1 + \frac{1}{r}\right)P\frac{dV}{d\eta} + \frac{1}{r}V\frac{dP}{d\eta}\right) + PV\left(\left(1 + \frac{1}{r}\right)c_1 + \frac{c_3}{r}\right)\right] \tag{321}$$

$$\left(kP - c^2\xi^2 V\right)\left(\frac{d^2V}{d\eta^2} - \frac{dV}{d\eta}\right) = c\left(c_1 + c\right)\xi^2 V\frac{dV}{d\eta} + \left(c_2 + c\right)\xi V\frac{dU}{d\eta} \tag{322}$$

$$- kqV^{-1}P\left(\frac{dV}{d\eta}\right)^2 - \left(n-1\right)VP^{-1}\left(\frac{dP}{d\eta}\right)^2 - \left(q + kn + 1\right)\frac{dP}{d\eta}\frac{dV}{d\eta} \tag{323}$$

$$+ \xi^2 V^{-q}P^{1-n}\left[c\left(\left(1 + \frac{1}{r}\right)P\frac{dV}{d\eta} + \frac{1}{r}V\frac{dP}{d\eta}\right) + PV\left(\left(1 + \frac{1}{r}\right)c_1 + \frac{c_3}{r}\right)\right] \tag{324}$$

pressure eq:

$$\frac{d^2P}{d\xi^2} = -c^2\xi^2\frac{d^2V}{d\xi^2} - c\left(c_1 + c\right)\xi V' - \left(c_2 + c\right)\frac{dU}{d\xi} \tag{325}$$

$$\frac{1}{\xi^2}\left(\frac{d^2P}{d\eta^2} - \frac{dP}{d\eta}\right) = -c^2\left(\frac{d^2V}{d\eta^2} - \frac{dV}{d\eta}\right) - c\left(c_1 + c\right)\frac{dV}{d\eta} - \left(c_2 + c\right)\frac{1}{\xi}\frac{dU}{d\eta} \tag{326}$$

$$\frac{d^2P}{d\eta^2} - \frac{dP}{d\eta} = -\xi^2c^2\left(\frac{d^2V}{d\eta^2} - \frac{dV}{d\eta}\right) - \xi^2c\left(c_1 + c\right)\frac{dV}{d\eta} - \left(c_2 + c\right)\xi\frac{dU}{d\eta} \tag{327}$$

1.4 working with $\ln\xi$ and $\ln f$

$$\frac{df}{d\ln x} = \frac{df}{dx}\frac{dx}{d\ln x} = x\frac{df}{dx} \tag{328}$$

$$\frac{d\ln f}{d\ln x} = x\frac{d\ln f}{dx} = \frac{x}{f}\frac{df}{dx} \tag{329}$$

$$\frac{d\ln f}{d\ln x} = \frac{1}{f}\frac{df}{d\ln x} \tag{330}$$

hence:

$$\frac{df}{dx} = \frac{1}{x}\frac{df}{d\ln x} \tag{331}$$

$$\frac{df}{dx} = \frac{f}{x}\frac{d\ln f}{d\ln x} \tag{332}$$

we saw:

$$\frac{d^2f}{dx^2} = \frac{1}{x^2}\left(\frac{d^2f}{d\left(\ln x\right)^2} - \frac{df}{d\ln x}\right) \tag{333}$$

now:

$$\frac{d^2 f}{dx^2} = \frac{1}{x^2}\left(\frac{d^2 f}{d\,(\ln x)^{\,2}} - \frac{df}{d\ln x}\right) \tag{334}$$

$$= \frac{1}{x^2}\left(\frac{d}{d\ln x}\left(\frac{df}{d\ln x}\right) - f\frac{d\ln f}{d\ln x}\right) \tag{335}$$

$$= \frac{1}{x^2}\left(\frac{d}{d\ln x}\left(f\frac{d\ln f}{d\ln x}\right) - f\frac{d\ln f}{d\ln x}\right) \tag{336}$$

$$= \frac{1}{x^2}\left(\frac{df}{d\ln x}\frac{d\ln f}{d\ln x} + f\frac{d^2\ln f}{d\,(\ln x)^{\,2}} - f\frac{d\ln f}{d\ln x}\right) \tag{337}$$

$$= \frac{1}{x^2}\left(f\frac{d\ln f}{d\ln x}\frac{d\ln f}{d\ln x} + f\frac{d^2\ln f}{d\,(\ln x)^{\,2}} - f\frac{d\ln f}{d\ln x}\right) \tag{338}$$

$$= \frac{f}{x^2}\left(\frac{d^2\ln f}{d\,(\ln x)^{\,2}} + \left(\frac{d\ln f}{d\ln x}\right)^2 - \frac{d\ln f}{d\ln x}\right) \tag{339}$$

summary:

$$\frac{df}{dx} = \frac{f}{x}\frac{d\ln f}{d\ln x} \tag{340}$$

$$\frac{d^2 f}{dx^2} = \frac{f}{x^2}\left(\frac{d^2\ln f}{d\,(\ln x)^{\,2}} + \left(\frac{d\ln f}{d\ln x}\right)^2 - \frac{d\ln f}{d\ln x}\right) \tag{341}$$

define

$$\eta = \ln\xi \tag{342}$$

eq are:

$$\frac{dU}{d\xi} = c\xi V' + c_1 V \tag{343}$$

which is

$$\frac{U}{\xi}\frac{d\ln U}{d\eta} = cV\frac{d\ln V}{d\eta} + c_1 V \tag{344}$$

$$\frac{d\ln U}{d\eta} = \frac{V\xi}{U}\left(c\frac{d\ln V}{d\eta} + c_1\right) \tag{345}$$

energy equation:

$$\left(kP - c^2\xi^2 V\right)\frac{d^2 V}{d\xi^2} = c\,(c_1 + c)\,\xi V V' + (c_2 + c)\,V\frac{dU}{d\xi} \tag{346}$$

$$-\,kqV^{-1}PV'^2 - (n-1)\,VP^{-1}P'^2 - (q + kn + 1)\,P'V' \tag{347}$$

$$+\,V^{-q}P^{1-n}\left[c\xi\left(\left(1 + \frac{1}{r}\right)PV' + \frac{1}{r}VP'\right) + PV\left(\left(1 + \frac{1}{r}\right)c_1 + \frac{c_3}{r}\right)\right] \tag{348}$$

xxxxxxxxxxxxxxxxx

$$\left(kP - c^2\xi^2 V\right)\frac{1}{\xi^2}\left(\frac{d^2V}{d\eta^2} - \frac{dV}{d\eta}\right) = c\left(c_1 + c\right)V\frac{dV}{d\eta} + \left(c_2 + c\right)V\frac{dU}{d\xi} \tag{349}$$

$$- kqV^{-1}PV'^2 - (n-1)\,VP^{-1}P'^2 - (q + kn + 1)\,P'V' \tag{350}$$

$$+ V^{-q}P^{1-n}\left[c\left(\left(1 + \frac{1}{r}\right)P\frac{dV}{d\eta} + \frac{1}{r}V\frac{dP}{d\eta}\right) + PV\left(\left(1 + \frac{1}{r}\right)c_1 + \frac{c_3}{r}\right)\right] \tag{351}$$

$$\left(kP - c^2\xi^2 V\right)\left(\frac{d^2V}{d\eta^2} - \frac{dV}{d\eta}\right) = c\left(c_1 + c\right)\xi^2 V\frac{dV}{d\eta} + \left(c_2 + c\right)\xi V\frac{dU}{d\eta} \tag{352}$$

$$- kqV^{-1}P\left(\frac{dV}{d\eta}\right)^2 - (n-1)\,VP^{-1}\left(\frac{dP}{d\eta}\right)^2 - (q + kn + 1)\frac{dP}{d\eta}\frac{dV}{d\eta} \tag{353}$$

$$+ \xi^2 V^{-q}P^{1-n}\left[c\left(\left(1 + \frac{1}{r}\right)P\frac{dV}{d\eta} + \frac{1}{r}V\frac{dP}{d\eta}\right) + PV\left(\left(1 + \frac{1}{r}\right)c_1 + \frac{c_3}{r}\right)\right] \tag{354}$$

pressure eq:

$$\frac{d^2P}{d\xi^2} = -c^2\xi^2\frac{d^2V}{d\xi^2} - c\left(c_1 + c\right)\xi V' - \left(c_2 + c\right)\frac{dU}{d\xi} \tag{355}$$

$$\frac{1}{\xi^2}\left(\frac{d^2P}{d\eta^2} - \frac{dP}{d\eta}\right) = -c^2\left(\frac{d^2V}{d\eta^2} - \frac{dV}{d\eta}\right) - c\left(c_1 + c\right)\frac{dV}{d\eta} - \left(c_2 + c\right)\frac{1}{\xi}\frac{dU}{d\eta} \tag{356}$$

$$\frac{d^2P}{d\eta^2} - \frac{dP}{d\eta} = -\xi^2 c^2\left(\frac{d^2V}{d\eta^2} - \frac{dV}{d\eta}\right) - \xi^2 c\left(c_1 + c\right)\frac{dV}{d\eta} - \left(c_2 + c\right)\xi\frac{dU}{d\eta} \tag{357}$$

1.5 constraints on the units from cancellation of unit terms

our equation was:

$$A^{a_1+a_3}B^{b_1+b_3}t^{c_1+c_3-1}\left[c\xi\left(\left(1 + \frac{1}{r}\right)PV' + \frac{1}{r}VP'\right) + PV\left(\left(1 + \frac{1}{r}\right)c_1 + \frac{c_3}{r}\right)\right] \tag{358}$$

$$= B\Bigg(v^{q-1}p^{n-2}\left(vp' + kpv'\right)\left(qpv' + (n-1)\,vp'\right) \tag{359}$$

$$+ p^{n-1}v^q\left[(k+1)\,v'p' + vp'' + kpv''\right]\Bigg) \tag{360}$$

using

$$\frac{\partial v}{\partial m} = A^{a_1+a}B^{b_1+b}t^{c_1+c}V' \tag{361}$$

$$\frac{\partial p}{\partial m} = A^{a_3+a}B^{b_3+b}t^{c_3+c}P' \tag{362}$$

$$v = VA^{a_1}B^{b_1}t^{c_1} \tag{363}$$

$$p = PA^{a_3}B^{b_3}t^{c_3} \tag{364}$$

hence unit factor of lhs

$$Bp^{n-1}v^q v' p' = B\left[A^{a_3(n-1)}B^{b_3(n-1)}t^{c_3(n-1)}\right]\left[A^{a_1 q}B^{b_1 q}t^{c_1 q}\right]\left[A^{a_1+a}B^{b_1+b}t^{c_1+c}\right]\left[A^{a_3+a}B^{b_3+b}t^{c_3+c}\right] \tag{365}$$

$$= \left[A^{a_1(q+1)+a_3 n+2a}B^{1+b_1(q+1)+b_3 n+2b}t^{c_1(q+1)+c_3 n+2c}\right] \tag{366}$$

hence we must have

$$A^{a_1+a_3}B^{b_1+b_3}t^{c_1+c_3-1} \tag{367}$$

$$= A^{a_1(q+1)+a_3 n+2a}B^{1+b_1(q+1)+b_3 n+2b}t^{c_1(q+1)+c_3 n+2c} \tag{368}$$

hence we must have:

$$t^{-1} = A^{a_1 q+a_3(n-1)+2a}B^{1+b_1 q+b_3(n-1)+2b}t^{c_1 q+c_3(n-1)+2c} \tag{369}$$

from which:

$$a_3 = -\frac{2a+a_1 q}{n-1} \tag{370}$$

$$b_3 = -\frac{1+2b+b_1 q}{n-1} \tag{371}$$

$$c_3 = -\frac{1+2c+c_1 q}{n-1} \tag{372}$$

2 Relation to Eulerian coordinate - positions in space

The ablated mass is:

$$\xi = mA^a B^b t^c \tag{373}$$

$$\boxed{m_f(t) = A^{-a}B^{-b}t^{-c}\xi_f} \tag{374}$$

which gives a condition on τ:

$$-c > 0 \tag{375}$$

$$c < 0 \tag{376}$$

$$\frac{3\mu - 2\lambda - 2 + \tau\left(2\left(\beta - \alpha - 4\right) - \beta\lambda + \mu\left(4+\alpha\right)\right)}{4 + 2\lambda - 4\mu} < 0 \tag{377}$$

we have the relations:

$$m(t,x) = \int_0^x \rho(x',t)\,dx' \tag{378}$$

$$x(t,m) - x_b(t) = \int_0^m \frac{dm'}{\rho(m',t)} = \int_0^m v(m',t)\, dm' \tag{379}$$

where $x_b(t)$ is the system boundary position (the system boundary on which the external temperature is applied).

2.1 Exact positions for given mass

distance from boundary as a function of mass:

$$x(t,m) - x_b(t) = \int_0^m v(m',t)\, dm' \tag{380}$$

since:

$$v(m,t) = V(\xi)\, A^{a_1} B^{b_1} t^{c_1} \tag{381}$$

$$x(t,m) - x_b(t) = \int_0^m v(m',t)\, dm' \tag{382}$$

$$= \int_0^m V(\xi(m'))\, A^{a_1} B^{b_1} t^{c_1} dm' \tag{383}$$

$$m' = A^{-a} B^{-b} t^{-c} \xi' \tag{384}$$

$$dm' = A^{-a} B^{-b} t^{-c} d\xi' \tag{385}$$

$$x(t,m) - x_b(t) = \int_0^{\xi(m)} V(\xi')\, A^{a_1} B^{b_1} t^{c_1} A^{-a} B^{-b} t^{-c} d\xi' \tag{386}$$

$$= A^{a_1 - a} B^{b_1 - b} t^{c_1 - c} \int_0^{\xi(m)} V(\xi')\, d\xi' \tag{387}$$

Hence:

$$x(t,m) - x_b(t) = A^{a_1 - a} B^{b_1 - b} t^{c_1 - c} \int_0^{\xi(m)} V(\xi')\, d\xi' \tag{388}$$

2.2 Distance boundary to ablation front

The distance from the system boundary on which the external temperature is applied to the heat front is:

$$\Delta x_f(t) = x(t, m_f) - x_b(t) \tag{389}$$

$$\Delta x_f(t) = A^{a_1 - a} B^{b_1 - b} t^{c_1 - c} \int_0^{\xi_f} V(\xi')\, d\xi' \tag{390}$$

so we need to calculate the area under $V(\xi)$.

2.3 Boundary position

$$u_b(t) \equiv u(m=0,t) \tag{391}$$

$$u(m,t) = U(\xi)\, A^{a_2} B^{b_2} t^{c_2} \tag{392}$$

$$u_b(t) \equiv U(0)\, A^{a_2} B^{b_2} t^{c_2} \tag{393}$$

$$x_b(t) = \int_0^t u_b(t')\, dt' = \int_0^t U(0)\, A^{a_2} B^{b_2} t'^{c_2} dt' \tag{394}$$

$$= U(0)\, A^{a_2} B^{b_2} \int_0^t t'^{c_2} dt' \tag{395}$$

$$= U(0)\, A^{a_2} B^{b_2} \frac{t^{c_2+1}}{c_2+1} \tag{396}$$

hence the boundary position is:

$$\boxed{x_b(t) = \frac{A^{a_2} B^{b_2}}{c_2+1} U(0)\, t^{c_2+1} \qquad (397)}$$

condition on τ due to the temporal integration?

$$c_2 > -1 \tag{398}$$

2.4 Ablation front position

we know that it must be zero:

$$x_f(t) \equiv x_b(t) + \Delta x_f(t) = 0 \tag{399}$$

since our boundary condition is:

$$u(m_f(t), t) = 0 \tag{400}$$

we will show this directly using $V(\xi)$ integral identity.

$$x_f(t) \equiv x_b(t) + \Delta x_f(t) \tag{401}$$

$$x_f(t) = \frac{A^{a_2} B^{b_2}}{c_2+1} U(0)\, t^{c_2+1} + A^{a_1-a} B^{b_1-b} t^{c_1-c} \int_0^{\xi_f} V(\xi')\, d\xi' \tag{402}$$

we already saw in the derivation of the self similar ODE for the mass equation that:

$$a_2 = a_1 - a,\ b_2 = b_1 - b,\ c_2 + 1 = c_1 - c \tag{403}$$

$$x_f\left(t\right)=A^{a_2}B^{b_2}t^{c_2+1}\left(\frac{U\left(0\right)}{c_2+1}+\int_0^{\xi_f}V\left(\xi'\right)d\xi'\right)=0 \tag{404}$$

we we have used the integral identity:

$$\int_0^{\xi_f}V\left(\xi'\right)d\xi'=-\frac{U\left(0\right)}{c_2+1} \tag{405}$$

which we will show later.

2.5 Ablation front velocity

$$v_f\left(t\right)=\frac{dx_f}{dt}=0 \tag{406}$$

2.6 Positions as a function of time

In ablated region:

$$x\left(t,m\right)-x_b\left(t\right)=\int_0^m\frac{dm'}{\rho\left(m',t\right)}=\int_0^m v\left(m',t\right)dm' \tag{407}$$

Hence:
for the ablated region:

$$\xi\left(m\right)=mA^aB^bt^c \tag{408}$$

$$x\left(t,m\right)=A^{a_2}B^{b_2}t^{c_2+1}\left(\frac{U\left(0\right)}{c_2+1}+\int_0^{\xi\left(m\right)}V\left(\xi'\right)d\xi'\right) \tag{409}$$

we will show in the next section the identity:

$$\int_0^{\xi_0}V\left(\xi\right)d\xi=\left(\frac{1}{c_2+1}\right)\left[U\left(\xi_0\right)-U\left(0\right)-c\xi_0V\left(\xi_0\right)\right] \tag{410}$$

hence:

$$x\left(t,m\right)=A^{a_2}B^{b_2}t^{c_2+1}\left(\frac{U\left(0\right)}{c_2+1}+\left(\frac{1}{c_2+1}\right)\left[U\left(\xi\left(m\right)\right)-U\left(0\right)-c\xi\left(m\right)V\left(\xi\left(m\right)\right)\right]\right) \tag{411}$$

Hence:

$$\boxed{\begin{aligned}\xi\left(m\right)&=mA^aB^bt^c &(412)\\ x\left(t,m\right)&=\frac{A^{a_2}B^{b_2}t^{c_2+1}}{c_2+1}\left[U\left(\xi\left(m\right)\right)-c\xi\left(m\right)V\left(\xi\left(m\right)\right)\right] &(413)\end{aligned}}$$

We can see that, since we get at the front $U_f = V_f = 0$, we should have at the front $x(t, m_f(t)) \propto U_f - c\xi_f V_f = 0$.

However, for any other value of m, **this is not always algebraically negative**, since $U \le 0, V \ge 0$ but $c < 0$.

In addition, it turns out that near $\xi \ge \xi_f$ we can get a small initial POSITIVE value of x which quickly turns negative.

Specifically, since in practice we do not really have $U_f = V_f = 0$ we can get a small positive number :(

so in practice, we look at the value of $U(\xi) - c\xi V(\xi)$ and if it is slightly positive we turn in to be 0. This way we will always have:

$$x(t, m \ge m_f(t)) = 0 \tag{414}$$

3 Calculation of $V(\xi)$ integral

the self similar mass equation is:

$$c\xi V' + c_1 V - U' = 0 \tag{415}$$

$$V(\xi) = \frac{1}{c_1}(U' - c\xi V') \tag{416}$$

$$V(\xi) = p\xi V' + qU' \tag{417}$$

$$p = -\frac{c}{c_1} \tag{418}$$

$$q = \frac{1}{c_1} \tag{419}$$

integration

$$\int_0^{\xi_0} V(\xi)\, d\xi = p\int_0^{\xi_0} \xi V'(\xi)\, d\xi + q(U(\xi_0) - U(0)) \tag{420}$$

integration by parts:

$$\int_0^{\xi_0} \xi V'(\xi)\, d\xi = [\xi V(\xi)]_0^{\xi_0} - \int_0^{\xi_0} V(\xi)\, d\xi \tag{421}$$

hence:

$$\int_0^{\xi_0} V(\xi)\, d\xi = p\left[[\xi V(\xi)]_0^{\xi_0} - \int_0^{\xi_0} V(\xi)\, d\xi\right] + q(U(\xi_0) - U(0)) \tag{422}$$

$$(1+p)\int_0^{\xi_0} V(\xi)\, d\xi = p[\xi V(\xi)]_0^{\xi_0} + q(U(\xi_0) - U(0)) \tag{423}$$

$$\int_0^{\xi_0} V(\xi)\, d\xi = \frac{p}{1+p}[\xi V(\xi)]_0^{\xi_0} + \frac{q}{1+p}(U(\xi_0) - U(0)) \tag{424}$$

$$q_1 = \frac{p}{1+p} = \frac{-\frac{c}{c_1}}{1-\frac{c}{c_1}} = -\frac{c}{c_1 - c} = \tag{425}$$

$$q_2 = \frac{q}{1+p} = \frac{\frac{1}{c_1}}{1-\frac{c}{c_1}} = \frac{1}{c_1 - c} \tag{426}$$

hence:

$$\int_0^{\xi_0} V(\xi)\, d\xi = \left(\frac{1}{c_1 - c}\right)\left[U(\xi_0) - U(0) - c\left[\xi V(\xi)\right]_0^{\xi_0}\right] \tag{427}$$

what is

$$\lim_{\xi \to 0}\left[V(\xi)\,\xi\right] \tag{428}$$

expand near zero

$$\lim_{\xi \to 0} V(\xi) \approx A\xi^k \tag{429}$$

since $\int_0^{\xi_0} V(\xi)\, d\xi$ represents the position of mass $m(\xi_0)$, it must be finite (for any m and hence ξ_0):

$$\int_0^{\xi_0} V(\xi)\, d\xi = \int_0^{\xi_0} A\xi^k d\xi \tag{430}$$

this integral is finite only if

$$k > -1 \tag{431}$$

or

$$k + 1 > 0 \tag{432}$$

hence

$$\lim_{\xi \to 0}\left[V(\xi)\,\xi\right] \approx \lim_{\xi \to 0}\left[A\xi^{k+1}\right] = 0 \tag{433}$$

hence:

$$\int_0^{\xi_0} V(\xi)\, d\xi = \left(\frac{1}{c_1 - c}\right)\left[U(\xi_0) - U(0) - c\xi_0 V(\xi_0)\right] \tag{434}$$

Summary

for any $\xi \leq \xi_f$:

$$\boxed{\int_0^{\xi} V(\xi')\, d\xi' = \left(\frac{1}{c_1 - c}\right)\left[U(\xi) - U(0) - c\xi V(\xi)\right]} \tag{435}$$

specifically for $\xi = \xi_f$:

$$\int_0^{\xi_f} V(\xi)\, d\xi = \left(\frac{1}{c_1 - c}\right)\left[U_f - U(0) - c\xi_f V_f\right] \tag{436}$$

we know that:

$$V_f = U_f = 0 \tag{437}$$

Hence:

$$\boxed{\int_0^{\xi_f} V(\xi)\, d\xi = -\frac{U(0)}{c_1 - c} = -\frac{U(0)}{c_2 + 1}} \tag{438}$$

positivity check:

$$\int_0^{\xi_f} V(\xi)\, d\xi = -\frac{U(0)}{c_1 - c} = \frac{-U(0)}{c_2 + 1} > 0 \tag{439}$$

since we already saw that:

$$c_2 + 1 = c_1 - c \tag{440}$$

$$c_2 > -1 \tag{441}$$

and we assume that the boundary is moving in the negative direction:

$$U(0) < 0 \tag{442}$$

4 Flux

$$\frac{\partial e}{\partial t} + p\frac{\partial v}{\partial t} = \left(\frac{\beta}{4+\alpha}\right) B \frac{\partial}{\partial m}\left(v^{\lambda}\frac{\partial}{\partial m}\left(pv^{1-\mu}\right)^{\frac{4+\alpha}{\beta}}\right) \tag{443}$$

$$\frac{\partial e}{\partial t} + p\frac{\partial v}{\partial t} = -\frac{\partial s}{\partial m} \tag{444}$$

$$s(m,t) = -\left(\frac{\beta}{4+\alpha}\right) B v^{\lambda}\frac{\partial}{\partial m}\left(pv^{1-\mu}\right)^{\frac{4+\alpha}{\beta}} \tag{445}$$

$$s(m,t) = -B\left(pv^{1-\mu}\right)^{\frac{4+\alpha}{\beta}-1} v^{\lambda}\frac{\partial}{\partial m}\left(pv^{1-\mu}\right) \tag{446}$$

$$n = \frac{4+\alpha}{\beta},\ \ k = 1-\mu,\ \ q = kn + \lambda - 1 \tag{447}$$

$$s(m,t) = -B\left(pv^{k}\right)^{n-1} v^{\lambda}\frac{\partial}{\partial m}\left(pv^{k}\right) \tag{448}$$

$$= -B\left(pv^{k}\right)^{n-1} v^{\lambda}\left(v^{k}p' + kpv^{k-1}v'\right) \tag{449}$$

$$= -Bp^{n-1}v^{k(n-1)+\lambda+k-1}\left(vp' + kpv'\right) \tag{450}$$

$$= -Bp^{n-1}v^{q}\left(vp' + kpv'\right) \tag{451}$$

$$v(m,t) = V(\xi)\, A^{a_1}B^{b_1}t^{c_1} \tag{452}$$

$$p(m,t) = P(\xi)\, A^{a_3}B^{b_3}t^{c_3} \tag{453}$$

$$\frac{\partial v}{\partial m} = A^{a_1+a}B^{b_1+b}t^{c_1+c}V' \tag{454}$$

$$\frac{\partial p}{\partial m} = A^{a_3+a}B^{b_3+b}t^{c_3+c}P' \tag{455}$$

$$s\left(m,t\right)=-Bp^{n-1}v^{q}\left(vp'+kpv'\right) \tag{456}$$

$$s\left(m,t\right)=-B\left(PA^{a_3}B^{b_3}t^{c_3}\right)^{n-1}\left(VA^{a_1}B^{b_1}t^{c_1}\right)^{q}\left(VA^{a_1}B^{b_1}t^{c_1}A^{a_3+a}B^{b_3+b}t^{c_3+c}P'+kPA^{a_3}B^{b_3}t^{c_3}A^{a_1+a}B^{b_1+b}t^{c_1} \tag{457}$$

$$=-A^{a_3(n-1)+qa_1+a_1+a_3+a}B^{1+b_3(n-1)+qb_1+b_1+b_3+b}t^{c_3(n-1)+qc_1+c_1+c_3+c}P^{n-1}V^{q}\left(VP'+kPV'\right) \tag{458}$$

$$=-A^{a+(q+1)a_1+na_3}B^{1+b+(q+1)b_1+nb_3}t^{c+(q+1)c_1+nc_3}P^{n-1}V^{q}\left(VP'+kPV'\right) \tag{459}$$

$$S\left(\xi\right)=-P^{n-1}V^{q}\left(VP'+kPV'\right)=-P^{n-1}V^{q-k+1}\left(V^{k}P\right)' \tag{460}$$

summary:

$$s\left(m,t\right)=A^{a+(q+1)a_1+na_3}B^{1+b+(q+1)b_1+nb_3}t^{c+(q+1)c_1+nc_3}S\left(\xi\right) \tag{461}$$

$$S\left(\xi\right)=-P^{n-1}\left(\xi\right)V^{q}\left(\xi\right)\left(V\left(\xi\right)P'\left(\xi\right)+kP\left(\xi\right)V'\left(\xi\right)\right) \tag{462}$$

5 Energy

internal energy:

$$E_{in}\left(t\right)=\int_{0}^{m_f(t)}e\left(m,t\right)dm=\frac{1}{r}\int_{0}^{m_f(t)}p\left(m,t\right)v\left(m,t\right)dm \tag{463}$$

$$v\left(m,t\right)=V\left(\xi\right)A^{a_1}B^{b_1}t^{c_1} \tag{464}$$

$$p\left(m,t\right)=P\left(\xi\right)A^{a_3}B^{b_3}t^{c_3} \tag{465}$$

$$\xi=mA^{a}B^{b}t^{c} \tag{466}$$

$$m=A^{-a}B^{-b}t^{-c}\xi \tag{467}$$

$$dm=A^{-a}B^{-b}t^{-c}d\xi \tag{468}$$

$$E_{in}\left(t\right)=\frac{1}{r}\int_{0}^{m_f(t)}p\left(m,t\right)v\left(m,t\right)dm \tag{469}$$

$$=\frac{1}{r}\int_{0}^{\xi_f}\left(P\left(\xi\right)A^{a_3}B^{b_3}t^{c_3}\right)\left(V\left(\xi\right)A^{a_1}B^{b_1}t^{c_1}\right)\left(A^{-a}B^{-b}t^{-c}\right)d\xi \tag{470}$$

so the internal energy:

$$E_{in}\left(t\right)=A^{a_3+a_1-a}B^{b_3+b_1-b}t^{c_3+c_1-c}\int_{0}^{\xi_f}\frac{P\left(\xi\right)V\left(\xi\right)}{r}d\xi \tag{471}$$

Kinetic energy:

$$E_k\left(t\right)=\int_{0}^{m_f(t)}\frac{1}{2}u^{2}\left(m,t\right)dm \tag{472}$$

$$u(m,t) = U(\xi) A^{a_2} B^{b_2} t^{c_2} \tag{473}$$

$$E_k(t) = \frac{1}{2} \int_0^{\xi_f} \left(U(\xi) A^{a_2} B^{b_2} t^{c_2}\right)^2 \left(A^{-a} B^{-b} t^{-c}\right) d\xi \tag{474}$$

so the kinetic energy:

$$E_k(t) = A^{2a_2-a} B^{2b_2-b} t^{2c_2-c} \int_0^{\xi_f} \frac{U^2(\xi)}{2} d\xi \tag{475}$$

we already saw the relation:

$$a_2 = a_1 - a,\ b_2 = b_1 - b,\ c_2 = c_1 - c - 1 \tag{476}$$

$$a_3 = a_1 - 2a,\ b_3 = b_1 - 2b,\ c_3 = c_1 - 2c - 2 \tag{477}$$

from which:

$$c_1 + c_3 = 2c_1 - 2c - 2 = 2(c_1 - c - 1) = 2c_2 \tag{478}$$

same for a, b:

$$a_1 + a_3 = 2a_2,\ b_1 + b_3 = 2b_2,\ c_1 + c_3 = 2c_2 \tag{479}$$

so E_{in}, E_k have the same powers.

5.1 An exact expression for total energy integral

Total energy equation:

$$\frac{\partial}{\partial t}\left(e + \frac{1}{2}u^2\right) + \frac{\partial(pu)}{\partial m} = \left(\frac{\beta}{4+\alpha}\right) B \frac{\partial}{\partial m}\left(v^\lambda \frac{\partial}{\partial m}\left(pv^{1-\mu}\right)^{\frac{4+\alpha}{\beta}}\right) \tag{480}$$

$$\frac{\text{rhs}}{B} = \left(\frac{\beta}{4+\alpha}\right) \frac{\partial}{\partial m}\left(v^\lambda \frac{\partial}{\partial m}\left(pv^{1-\mu}\right)^{\frac{4+\alpha}{\beta}}\right) \tag{481}$$

$$= \frac{1}{n}\frac{\partial}{\partial m}\left(v^\lambda \frac{\partial}{\partial m}\left(pv^k\right)^n\right) = \frac{1}{n} n \frac{\partial}{\partial m}\left(v^\lambda \left(pv^k\right)^{n-1} \frac{\partial}{\partial m}\left(pv^k\right)\right) \tag{482}$$

$$= \left(v^\lambda \left(pv^k\right)^{n-1} \left(pv^k\right)'\right)' \tag{483}$$

$$\frac{\partial}{\partial t}\left(\frac{pv}{r} + \frac{1}{2}u^2\right) + (pu)' = \left(v^\lambda \left(pv^k\right)^{n-1} \left(pv^k\right)'\right)' \tag{484}$$

$$\frac{\partial v}{\partial t} = A^{a_1} B^{b_1} t^{c_1-1} \left(c\xi V' + c_1 V\right) \tag{485}$$

$$\frac{\partial u}{\partial t} = A^{a_2} B^{b_2} t^{c_2 - 1} \left(c\xi U' + c_2 U \right) \tag{486}$$

$$\frac{\partial p}{\partial t} = A^{a_3} B^{b_3} t^{c_3 - 1} \left(c\xi P' + c_3 P \right) \tag{487}$$

$$\frac{\partial u}{\partial m} = A^{a_2 + a} B^{b_2 + b} t^{c_2 + c} U' \tag{488}$$

$$\frac{\partial p}{\partial m} = A^{a_3 + a} B^{b_3 + b} t^{c_3 + c} P' \tag{489}$$

assuming all units drop:

$$\begin{aligned} \text{lhs} &= \frac{1}{r} \left(\left(c\xi V' + c_1 V \right) P + \left(c\xi P' + c_3 P \right) V \right) + \left(c\xi U' + c_2 U \right) U + \left(PU \right)' && (490) \\ &= \left(c_1 + c_3 \right) \frac{PV}{r} + c_2 U^2 + \frac{c}{r} \xi \left(V' P + V P' \right) + \frac{1}{2} c\xi \left(U^2 \right)' + \left(PU \right)' && (491) \end{aligned}$$

we saw that:

$$c_1 + c_3 = 2c_2 \tag{492}$$

$$\text{lhs} = 2c_2 \left(\frac{PV}{r} + \frac{1}{2} U^2 \right) + c\xi \left(\frac{PV}{r} + \frac{1}{2} U^2 \right)' + \left(PU \right)' \tag{493}$$

the equation is:

$$2c_2 \left(\frac{PV}{r} + \frac{1}{2} U^2 \right) + c\xi \left(\frac{PV}{r} + \frac{1}{2} U^2 \right)' + \left(PU \right)' = \left(V^{\lambda} \left(PV^k \right)^{n-1} \left(PV^k \right)' \right)' \tag{494}$$

$$2c_2 \int_0^{\xi_0} \left(\frac{PV}{r} + \frac{1}{2} U^2 \right) d\xi + c \int_0^{\xi_0} \xi \left(\frac{PV}{r} + \frac{1}{2} U^2 \right)' d\xi = \left[V^{\lambda} \left(PV^k \right)^{n-1} \left(PV^k \right)' - PU \right]_0^{\xi_0} \tag{495}$$

$$\int_0^{\xi_0} \xi f' \left(\xi \right) d\xi = \left[\xi f \left(\xi \right) \right]_0^{\xi_0} - \int_0^{\xi_0} f \left(\xi \right) d\xi \tag{496}$$

$$2c_2 \int_0^{\xi_0} \left(\frac{PV}{r} + \frac{1}{2} U^2 \right) d\xi + c \left(\left[\xi \frac{PV}{r} + \frac{1}{2} \xi U^2 \right]_0^{\xi_0} - \int_0^{\xi_0} \left(\frac{PV}{r} + \frac{1}{2} U^2 \right) d\xi \right) = \left[V^{\lambda} \left(PV^k \right)^{n-1} \left(PV^k \right)' - PU \right]_0^{\xi_0} \tag{497}$$

$$\left(2c_2 - c \right) \int_0^{\xi_0} \left(\frac{PV}{r} + \frac{1}{2} U^2 \right) d\xi = \left[V^{\lambda} \left(PV^k \right)^{n-1} \left(PV^k \right)' - PU - c\xi \left(\frac{PV}{r} + \frac{1}{2} U^2 \right) \right]_0^{\xi_0} \tag{498}$$

$$\int_0^{\xi_0} \left(\frac{PV}{r} + \frac{1}{2} U^2 \right) d\xi = \frac{1}{2c_2 - c} \left[V^{\lambda} \left(PV^k \right)^{n-1} \left(PV^k \right)' - PU - c\xi \left(\frac{PV}{r} + \frac{1}{2} U^2 \right) \right]_0^{\xi_0} \tag{499}$$

hence:

$$\int_0^{\xi_0}\left(\frac{PV}{r}+\frac{1}{2}U^2\right)d\xi=\frac{1}{2c_2-c}\left[V^{\lambda}\left(PV^k\right)^{n-1}\left(V^kP'+kPV^{k-1}V'\right)-PU-c\xi\left(\frac{PV}{r}+\frac{1}{2}U^2\right)\right]_0^{\xi_0} \tag{500}$$

Summary:

internal energy:

$$E_{in}\left(t\right)=A^{2a_2-a}B^{2b_2-b}t^{2c_2-c}\int_0^{\xi_f}\frac{P\left(\xi\right)V\left(\xi\right)}{r}d\xi \tag{501}$$

kinetic energy:

$$E_k\left(t\right)=A^{2a_2-a}B^{2b_2-b}t^{2c_2-c}\int_0^{\xi_f}\frac{U^2\left(\xi\right)}{2}d\xi \tag{502}$$

we have the identity:

$$\frac{E_k\left(t\right)}{E_{in}\left(t\right)}=const \tag{503}$$

total energy:

$$E\left(t\right)=E_{in}\left(t\right)+E_k\left(t\right)=A^{2a_2-a}B^{2b_2-b}t^{2c_2-c}\int_0^{\xi_f}\left(\frac{P\left(\xi\right)V\left(\xi\right)}{r}+\frac{U^2\left(\xi\right)}{2}\right)d\xi \tag{504}$$

total energy integral identity for every $\xi_0\leq\xi_f$:

$$\int_0^{\xi_0}\left(\frac{P\left(\xi\right)V\left(\xi\right)}{r}+\frac{1}{2}U^2\left(\xi\right)\right)d\xi=\frac{1}{2c_2-c}\left[V^{\lambda}\left(PV^k\right)^{n-1}\left(V^kP'+kPV^{k-1}V'\right)-PU-c\xi\left(\frac{PV}{r}+\frac{1}{2}U^2\right)\right]_0^{\xi_0} \tag{505}$$

6 Asymptotic analysis $\xi\to\xi_f$

assuming that for $\xi\to\xi_f$:

$$V\left(\xi\right)\approx V^*\left(\xi_f-\xi\right)^{\delta},\ 0<\delta<1 \tag{506}$$

Since we have the b.c. $V\left(\xi_f\right)=0$.
plugging into mass equation:

$$c\xi V'+c_1V-U'=0 \tag{507}$$

$$c\xi\left(-\delta V^*\left(\xi_f-\xi\right)^{\delta-1}\right)+c_1V^*\left(\xi_f-\xi\right)^{\delta}-U'=0 \tag{508}$$

using the approximations

$$\xi\left(\xi_f-\xi\right)^{\delta-1}\approx\xi_f\left(\xi_f-\xi\right)^{\delta-1} \tag{509}$$

$$\left(\xi_f-\xi\right)^{\delta-1} \gg \left(\xi_f-\xi\right)^{\delta} \tag{510}$$

we find:

$$U' = -c\xi_f\delta V^*\left(\xi_f-\xi\right)^{\delta-1} \tag{511}$$

integrating:

$$U\left(\xi_f\right)-U\left(\xi\right) = \left[c\xi_f V^*\left(\xi_f-\xi'\right)^{\delta}\right]_{\xi}^{\xi_f} \tag{512}$$

$$= -c\xi_f V^*\left(\xi_f-\xi'\right)^{\delta} \tag{513}$$

using the b.c. $U\left(\xi_f\right)=0$:

$$U\left(\xi\right) = c\xi_f V^*\left(\xi_f-\xi\right)^{\delta} \tag{514}$$

$$U\left(\xi\right) = c\xi_f V\left(\xi\right) \tag{515}$$

the momentum equation:

$$c\xi U' + c_2 U + P' = 0 \tag{516}$$

$$c\xi\left(-\delta c\xi_f V^*\left(\xi_f-\xi\right)^{\delta-1}\right) + c_2\left(c\xi_f V^*\left(\xi_f-\xi\right)^{\delta}\right) + P' = 0 \tag{517}$$

using the same approximations as before:

$$-\delta c^2\xi_f^2 V^*\left(\xi_f-\xi\right)^{\delta-1} + P' = 0 \tag{518}$$

$$P' = \delta c^2\xi_f^2 V^*\left(\xi_f-\xi\right)^{\delta-1} \tag{519}$$

integrating:

$$P\left(\xi_f\right)-P\left(\xi\right) = P_f - P\left(\xi\right) = \left[-c^2\xi_f^2 V^*\left(\xi_f-\xi'\right)^{\delta}\right]_{\xi}^{\xi_f} \tag{520}$$

$$= c^2\xi_f^2 V^*\left(\xi_f-\xi\right)^{\delta} \tag{521}$$

$$P\left(\xi\right) = P_f - c^2\xi_f^2 V^*\left(\xi_f-\xi\right)^{\delta} \tag{522}$$

$$P\left(\xi\right) = P_f - c^2\xi_f^2 V\left(\xi\right) \tag{523}$$

the energy equation:

$$c\xi\left(\left(1+\frac{1}{r}\right)PV' + \frac{1}{r}VP'\right) + PV\left(\left(1+\frac{1}{r}\right)c_1 + \frac{c_3}{r}\right) = \left(V^{\lambda}\left(PV^k\right)^{n-1}\left(PV^k\right)'\right)' \tag{524}$$

using

$$P\left(\xi\right) = P_f - c^2\xi_f^2 V\left(\xi\right) \tag{525}$$

we see that:

$$PV^k \approx P_f V^k \tag{526}$$

$$\begin{aligned}
\text{rhs} &= \left(V^\lambda \left(PV^k\right)^{n-1} \left(PV^k\right)'\right)' && (527)\\
&= \left(V^\lambda \left(P_f V^k\right)^{n-1} \left(P_f V^k\right)'\right)' && (528)\\
&= P_f^n \left(V^{\lambda+k(n-1)} \left(V^k\right)'\right)' && (529)\\
&= kP_f^n \left(V^{\lambda+k(n-1)} V^{k-1} V'\right)' && (530)\\
&= kP_f^n \left(V^q V'\right)' && (531)
\end{aligned}$$

since

$$\begin{aligned}
V(\xi) &= V^* (\xi_f - \xi)^\delta && (532)\\
V'(\xi) &= -\delta V^* (\xi_f - \xi)^{\delta-1} && (533)\\
V^q V' &= -\delta V^{*q+1} (\xi_f - \xi)^{\delta(q+1)-1} && (534)\\
\left(V^q V'\right)' &= (\delta(q+1) - 1)\, \delta V^{*q+1} (\xi_f - \xi)^{\delta(q+1)-2} && (535)
\end{aligned}$$

$$\begin{aligned}
\text{rhs} &= kP_f^n \left(V^q V'\right)' && (536)\\
&= kP_f^n (\delta(q+1) - 1)\, \delta V^{*q+1} (\xi_f - \xi)^{\delta(q+1)-2} && (537)
\end{aligned}$$

for the lhs we note that:

$$\begin{aligned}
PV &\approx P_f V && (538)\\
PV' &= P_f V' && (539)\\
VP' &= -c^2 \xi_f^2 V^2 && (540)
\end{aligned}$$

$$\text{lhs} = c\xi \left(\left(1 + \frac{1}{r}\right) PV' + \frac{1}{r} VP'\right) + PV \left(\left(1 + \frac{1}{r}\right) c_1 + \frac{c_3}{r}\right) \tag{541}$$

PV' has the lowest power of $\xi_f - \xi$:

$$\begin{aligned}
\text{lhs} &= c\xi \left(1 + \frac{1}{r}\right) PV' && (542)\\
&= c\xi \left(1 + \frac{1}{r}\right) P_f V' && (543)\\
&= c\xi \left(1 + \frac{1}{r}\right) P_f \left(-\delta V^* (\xi_f - \xi)^{\delta-1}\right) && (544)\\
&= -\left(1 + \frac{1}{r}\right) \delta c \xi_f P_f V^* (\xi_f - \xi)^{\delta-1} && (545)
\end{aligned}$$

so we have the equation:

$$-\left(1+\frac{1}{r}\right)\delta c\xi_f P_f V^*\left(\xi_f-\xi\right)^{\delta-1}=kP_f^n\left(\delta\left(q+1\right)-1\right)\delta V^{*q+1}\left(\xi_f-\xi\right)^{\delta(q+1)-2} \tag{546}$$

which means:

$$\delta-1=\delta\left(q+1\right)-2 \tag{547}$$

$$1=\delta q \tag{548}$$

$$\delta=\frac{1}{q} \tag{549}$$

$$-\left(1+\frac{1}{r}\right)\delta c\xi_f P_f V^*\left(\xi_f-\xi\right)^{\delta-1}=kP_f^n\left(\delta\left(q+1\right)-1\right)\delta V^{*q+1}\left(\xi_f-\xi\right)^{\delta(q+1)-2} \tag{550}$$

$$-\left(1+\frac{1}{r}\right)c\xi_f P_f V^*=kP_f^n\left(\delta\left(q+1\right)-1\right)V^{*q+1} \tag{551}$$

$$-\left(1+\frac{1}{r}\right)c\xi_f P_f^{1-n}=k\left(\frac{1}{q}\left(q+1\right)-1\right)V^{*q} \tag{552}$$

$$-\left(1+\frac{1}{r}\right)c\xi_f P_f^{1-n}=\frac{k}{q}V^{*q} \tag{553}$$

$$V^*=\left(-\left(1+\frac{1}{r}\right)\frac{qc}{k}\xi_f P_f^{1-n}\right)^{\frac{1}{q}} \tag{554}$$

SUMMARY:
near $\xi\to\xi_f$:

$$V\left(\xi\right)\approx V^*\left(\xi_f-\xi\right)^{\frac{1}{q}}, \tag{555}$$

$$U\left(\xi\right)\approx c\xi_f V\left(\xi\right) \tag{556}$$

$$P\left(\xi\right)\approx P_f-c^2\xi_f^2 V\left(\xi\right) \tag{557}$$

$$V^*=\left(-\left(1+\frac{1}{r}\right)\frac{qc}{k}\xi_f P_f^{1-n}\right)^{\frac{1}{q}} \tag{558}$$

$$q=kn+\lambda-1 \tag{559}$$

where

$$n=\frac{4+\alpha}{\beta},\ k=1-\mu \tag{560}$$

I think this matches Pakula-Sigel equations 20-22. Pakula-Sigel has our $\mu=0$ and hence $k=1$ and

$$q=kn+\lambda-1=n+\lambda-1 \tag{561}$$

$$n = \frac{4+\alpha}{\beta}, \tag{562}$$

$$n - 1 = \frac{4+\alpha-\beta}{\beta}, \tag{563}$$

7 Asymptotic analysis $\xi \to 0$

assuming

$$V(\xi) = V_0 \xi^w \tag{564}$$

$$w < 0 \tag{565}$$

mass equation:

$$c\xi V' + c_1 V - U' = 0 \tag{566}$$

$$c\xi V_0 w \xi^{w-1} + c_1 V_0 \xi^w - U' = 0 \tag{567}$$

$$U' = V_0 (cw + c_1) \xi^w \tag{568}$$

$$U(\xi) = U_0 + \frac{V_0 (cw + c_1)}{w+1} \xi^{w+1} \tag{569}$$

$$U(\xi) = U_0 + U_1 \xi^{w+1} \tag{570}$$

$$U_1 = \frac{V_0 (cw + c_1)}{w+1} \tag{571}$$

$$w + 1 > 0 \tag{572}$$

putting this in momentum equation:

$$c\xi U' + c_2 U + P' = 0 \tag{573}$$

$$c\xi (w+1) U_1 \xi^w + c_2 \left(U_0 + U_1 \xi^{w+1}\right) + P' = 0 \tag{574}$$

$$P' = -c_2 U_0 - (c(w+1) + c_2) U_1 \xi^{w+1} \tag{575}$$

$$P = P_0 - c_2 U_0 \xi - \frac{c(w+1) + c_2}{w+2} U_1 \xi^{w+2} \tag{576}$$

$$P = P_0 + P_1 \xi + P_2 \xi^{w+2} \tag{577}$$

the energy equation:

$$c\xi \left(\left(1 + \frac{1}{r}\right) PV' + \frac{1}{r} VP'\right) + PV \left(\left(1 + \frac{1}{r}\right) c_1 + \frac{c_3}{r}\right) = \left(V^\lambda \left(PV^k\right)^{n-1} \left(PV^k\right)'\right)' \tag{578}$$

$$V(\xi) = V_0 \xi^w \tag{579}$$

$$P = P_0 + P_1 \xi + P_2 \xi^{w+2} \tag{580}$$

XXXXXXXXXXXXXXX

8 Henyey like?

$$\xi = m\left(A^{2\beta-\beta\lambda-8-2\alpha+\mu(4+\alpha)}B^{\mu-2}t^{3\mu-2\lambda-2+\tau(2(\beta-\alpha-4)-\beta\lambda+\mu(4+\alpha))}\right)^{\frac{1}{4+2\lambda-4\mu}} \tag{581}$$

$$\frac{3\mu-2\lambda-2+\tau\left(2\left(\beta-\alpha-4\right)-\beta\lambda+\mu\left(4+\alpha\right)\right)}{4+2\lambda-4\mu} = -1 \tag{582}$$

$$3\mu-2\lambda-2+\tau\left(2\left(\beta-\alpha-4\right)-\beta\lambda+\mu\left(4+\alpha\right)\right) = 4\mu-4-2\lambda \tag{583}$$

$$\tau\left(2\left(\beta-\alpha-4\right)-\beta\lambda+\mu\left(4+\alpha\right)\right) = \mu-2 \tag{584}$$

$$\tau = \frac{\mu-2}{2\left(\beta-\alpha-4\right)-\beta\lambda+\mu\left(4+\alpha\right)} \tag{585}$$

9 Grand summary

The one temperature rad-hydro equations are:

$$\frac{\partial v}{\partial t} - \frac{\partial u}{\partial m} = 0 \tag{586}$$

$$\frac{\partial u}{\partial t} + \frac{\partial p}{\partial m} = 0 \tag{587}$$

$$\frac{\partial e}{\partial t} + p\frac{\partial v}{\partial t} = -\frac{\partial s}{\partial m} \tag{588}$$

where the radiation energy flux (energy per unit area per unit time):

$$s\left(m,t\right) = -\frac{c_{\text{light}}}{3\kappa_R}\frac{\partial}{\partial m}\left(aT^4\right) \tag{589}$$

where $v = 1/\rho$ is the specific volume and κ_R is the Rosseland opacity.
Assuming a material with the following properties:
an ideal gas law:

$$pv = \left(\gamma-1\right)e = re \tag{590}$$

a power law Rosseland opacity:

$$\frac{1}{\kappa_R\left(T,\rho\right)} = gT^{\alpha}\rho^{-\lambda} = gT^{\alpha}v^{\lambda} \tag{591}$$

and power law specific internal energy (energy per unit mass):

$$e\left(T,\rho\right) = fT^{\beta}\rho^{-\mu} = fT^{\beta}v^{\mu} \tag{592}$$

which gives the realtion for the temperature:

$$T\left(m,t\right) = \left(\frac{p\left(x,t\right)v^{1-\mu}\left(x,t\right)}{rf}\right)^{\frac{1}{\beta}} \tag{593}$$

The energy flux for this power-law opacity:

$$s(m,t) = -\frac{B}{n} v^{\lambda} \frac{\partial}{\partial m}\left(\left(pv^{k}\right)^{n}\right) = -Bv^{\lambda}\left(pv^{k}\right)^{n-1}\frac{\partial}{\partial m}\left(pv^{k}\right) \tag{594}$$

$$n = \frac{4+\alpha}{\beta},\ k = 1-\mu \tag{595}$$

The boundary conditions at the origin are:

$$T(0,t) = T_0 t^{\tau},\ p(0,t) = 0 \tag{596}$$

and at the ablation front:

$$\rho\left(m_f(t), t\right) = \infty \tag{597}$$

Under these assumptions the temperature can be eliminated from the problem. The energy equation can be written as:

$$\frac{\partial e}{\partial t} + p\frac{\partial v}{\partial t} = -\frac{\partial s}{\partial m}, \tag{598}$$

$$\frac{\partial e}{\partial t} + p\frac{\partial v}{\partial t} = B\frac{\partial}{\partial m}\left(v^{\lambda}\left(pv^{k}\right)^{n-1}\frac{\partial}{\partial m}\left(pv^{k}\right)\right), \tag{599}$$

and the boundary conditions read:

$$p(0,t)\, v^{1-\mu}(0,t) = A^{\beta} t^{\tau\beta},\ p(0,t) = 0. \tag{600}$$

We assume that there is an ablation front $m_f(t)$ on which:

$$v\left(m_f(t), t\right) = 0,\ u\left(m_f(t), t\right) = 0. \tag{601}$$

This problem is characterized by two dimensional parameters:

$$A = T_0 (rf)^{\frac{1}{\beta}},\ B = \frac{16\sigma g}{3\beta (rf)^{\frac{4+\alpha}{\beta}}} \tag{602}$$

A self similar solution exists:

$$\xi = m\left(A^{2\beta-\beta\lambda-8-2\alpha+\mu(4+\alpha)} B^{\mu-2} t^{3\mu-2\lambda-2+\tau(2(\beta-\alpha-4)-\beta\lambda+\mu(4+\alpha))}\right)^{\frac{1}{4+2\lambda-4\mu}} \tag{603}$$

$$v(m,t) = V(\xi)\left(A^{2\beta-4-\alpha} B^{-1} t^{1-\tau(4+\alpha-2\beta)}\right)^{\frac{1}{2+\lambda-2\mu}} \tag{604}$$

$$u(m,t) = U(\xi)\left(A^{\beta(2+\lambda)-\mu(4+\alpha)} B^{-\mu} t^{\tau(\beta(2+\lambda)-\mu(4+\alpha))+\mu}\right)^{\frac{1}{4+2\lambda-4\mu}} \tag{605}$$

$$p(m,t) = P(\xi)\left(A^{\beta\lambda+4+\alpha-\mu(4+\alpha)} B^{1-\mu} t^{\tau(\beta\lambda+4+\alpha-\mu(4+\alpha))+\mu-1}\right)^{\frac{1}{2+\lambda-2\mu}} \tag{606}$$

which we re-write as:

$$\xi = mA^a B^b t^c \tag{607}$$

$$v\left(m,t\right) = V\left(\xi\right) A^{a_1} B^{b_1} t^{c_1} \tag{608}$$

$$u\left(m,t\right) = U\left(\xi\right) A^{a_2} B^{b_2} t^{c_2} \tag{609}$$

$$p\left(m,t\right) = P\left(\xi\right) A^{a_3} B^{b_3} t^{c_3} \tag{610}$$

for convenience we also define a self similar temperature:

$$\mathcal{T}\left(\xi\right) \equiv \left(P\left(\xi\right) V\left(\xi\right)^{1-\mu}\right)^{\frac{1}{\beta}} \tag{611}$$

The self similar mass, momentum and energy ODEs are:

$$c\xi V' + c_1 V - U' = 0 \tag{612}$$

$$c\xi U' + c_2 U + P' = 0 \tag{613}$$

$$\begin{aligned} kPV'' + VP'' = &-kqV^{-1}PV'^2 - \left(n-1\right)VP^{-1}P'^2 - \left(q+kn+1\right)P'V' \\ &+ V^{-q}P^{1-n}\left(c\xi\left[\left(1+\frac{1}{r}\right)PV' + \frac{1}{r}VP'\right] + PV\left[\left(1+\frac{1}{r}\right)c_1 + \frac{c_3}{r}\right]\right) \end{aligned} \tag{614}$$

where:

$$q = kn + \lambda - 1 \tag{615}$$

In order to solve this numerically, we cast it into a first order ODE in a canonical form $\frac{d\boldsymbol{y}}{d\xi} = \boldsymbol{F}\left(\boldsymbol{y},\xi\right)$:

$$\frac{dV}{d\xi} = V' \tag{616}$$

$$\frac{dP}{d\xi} = P' \tag{617}$$

$$\frac{dU}{d\xi} = c\xi V' + c_1 V \tag{618}$$

$$\begin{aligned} \left(kP - c^2\xi^2 V\right)\frac{d^2V}{d\xi^2} = &\; c\left(c_1+c\right)\xi VV' + \left(c_2+c\right)V\frac{dU}{d\xi} \\ &- kqV^{-1}PV'^2 - \left(n-1\right)VP^{-1}P'^2 - \left(q+kn+1\right)P'V' \\ &+ V^{-q}P^{1-n}\left[c\xi\left(\left(1+\frac{1}{r}\right)PV' + \frac{1}{r}VP'\right) + PV\left(\left(1+\frac{1}{r}\right)c_1 + \frac{c_3}{r}\right)\right] \end{aligned} \tag{619}$$

$$\frac{d^2P}{d\xi^2} = -c^2\xi^2\frac{d^2V}{d\xi^2} - c\left(c_1+c\right)\xi V' - \left(c_2+c\right)\frac{dU}{d\xi} \tag{620}$$

written in terms of 5 variables $\boldsymbol{y} = (V, V', P, P', U)$ and 5 equations for their derivatives $\frac{d\boldsymbol{y}}{d\xi} = \left(\frac{dV}{d\xi}, \frac{d^2V}{d\xi^2}, \frac{dP}{d\xi}, \frac{d^2P}{d\xi^2}, \frac{dU}{d\xi}\right)$.

The boundary conditions for the self similar solution are:

$$V(\xi_f) = V'(\xi_f) = U(\xi_f) = 0 \tag{621}$$

$$P(0) = 0 \tag{622}$$

$$\mathcal{T}(0) = \left(P(0)\,V(0)^{1-\mu}\right)^{\frac{1}{\beta}} = 1 \tag{623}$$

which is solved by a double shooting method - for each ξ_f finding which P_f we get $P(0) = 0$ by integrating inwards from ξ_f with initial value of P_f, and then finding for which ξ_f this process also yields $\mathcal{T}(0) = 1$. In practice, in order to obtain a good estimate for the initial conditions for the numerical integration $\boldsymbol{y}(\xi_f)$, we use the asymptotic solution near the front $\xi \to \xi_f$, which is given by:

$$V(\xi) \approx V^*(\xi_f - \xi)^{\frac{1}{q}}, \tag{624}$$

$$U(\xi) \approx c\xi_f V(\xi) \tag{625}$$

$$P(\xi) \approx P_f - c^2\xi_f^2 V(\xi) \tag{626}$$

where:

$$V^* = \left(-\left(1 + \frac{1}{r}\right)\frac{qc}{k}\xi_f P_f^{1-n}\right)^{\frac{1}{q}} \tag{627}$$

The ablated mass is:

$$m_f(t) = A^{-a}B^{-b}t^{-c}\xi_f \tag{628}$$

which gives a condition (we want the heat wave to propagate outwards):

$$c < 0 \tag{629}$$

The boundary position is given by:

$$x_b(t) = \frac{A^{a_2}B^{b_2}U(0)}{c_2+1}t^{c_2+1} \tag{630}$$

The position of an element with Lagrangian coordinate m, as a function of time is given by:

$$x(t,m) = \begin{cases} \frac{A^{a_2}B^{b_2}t^{c_2+1}}{c_2+1}\left[U(\xi(m)) - c\xi(m)\,V(\xi(m))\right] & m < m_f(t) \\ 0 & \text{else} \end{cases} \tag{631}$$

where $\xi(m)$ is given by eq. 607.

Internal energy:

$$E_{in}(t) = A^{2a_2-a}B^{2b_2-b}t^{2c_2-c}\int_0^{\xi_f}\frac{P(\xi)V(\xi)}{r}d\xi \tag{632}$$

kinetic energy:

$$E_k(t) = A^{2a_2-a}B^{2b_2-b}t^{2c_2-c}\int_0^{\xi_f}\frac{U^2(\xi)}{2}d\xi \tag{633}$$

we have the identity:

$$\frac{E_k(t)}{E_{in}(t)} = const \tag{634}$$

total energy:

$$E(t) = E_{in}(t) + E_k(t) = A^{2a_2-a}B^{2b_2-b}t^{2c_2-c}\int_0^{\xi_f}\left(\frac{P(\xi)V(\xi)}{r} + \frac{U^2(\xi)}{2}\right)d\xi \tag{635}$$

we have the exact identity for any $\xi_0 \leq \xi_f$:

$$\int_0^{\xi_0}\left(\frac{PV}{r} + \frac{1}{2}U^2\right)d\xi = \frac{1}{2c_2-c}\left[-S(\xi) - P(\xi)U(\xi) - c\xi\left(\frac{P(\xi)V(\xi)}{r} + \frac{1}{2}U^2(\xi)\right)\right]_0^{\xi_0} \tag{636}$$

where the self similar flux profile is:

$$\begin{aligned} S(\xi) &= -\frac{1}{n}V^{\lambda}\left(\left(PV^k\right)^n\right)' = -V^{\lambda}\left(PV^k\right)^{n-1}\left(PV^k\right)' \\ &= -P^{n-1}V^q\left(VP' + kPV'\right) \end{aligned} \tag{637}$$

the dimensional radiation energy flux is given by:

$$s(m,t) = A^{a+(q+1)a_1+na_3}B^{1+b+(q+1)b_1+nb_3}t^{c+(q+1)c_1+nc_3}S(\xi) \tag{638}$$

10 Stitching subsonic heat wave and the shock

the pressure at the origin of the shock problem is given from the subsonic heat wave solution as $P_f A^{a_3}B^{b_3}t^{c_3}$ hence:

$$p_{shock}(0,t) = p_{0,s}t^{\tau_s} \tag{639}$$

with

$$p_{0,s} = P_f A^{a_3}B^{b_3}, \quad \tau_s = c_3 \tag{640}$$

Heat position is given by:

$$x_f(t) \equiv x^{shock}(t, m_f(t)) \tag{641}$$

where $x^{shock}(t,m)$ is the position of a Lagrangian coordinate m at time t, according to the piston shock problem solution (that is, it is somewhere within the piston and the shock front). Any other position in the ablative heat region will be re-positioned by $x_f(t)$ (the ablation front). Hence, instead of having $x_f(t) \equiv 0$ as in the pure ablation problem which assume in infinite wall at $x=0$, we have an inward moving front from which the ablated masses are coming out of. Therefore, the positions in the entire problem are given by:

$$x(t,m) = \begin{cases} x^{heat}(t,m) + x_f(t) & m < m_f(t) \\ x^{shock}(t,m) & \text{else} \end{cases} \tag{642}$$

where $x^{heat}(t,m)$ is the position of a Lagrangian element, according to the subsonic ablation heat wave solution.

For the hydrodynamic profiles $h(m,t) = v, u, p$:

$$h(m,t) = \begin{cases} h^{heat}(m,t) & m < m_f(t) \\ h^{shock}(m,t) & \text{else} \end{cases} \tag{643}$$